MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: CALCULATION

3340 CONCENTRATE THICKENING

DESIGN CALCULATION

AS BUILT

MD-216-3340-EG-CV-CAL-1220

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Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
API 650, 12th Edition	Welded Tanks For Oil Storage

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1 INTENT

This document is prepared for design of Concentrate Filtration (3340) Tank Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

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2 SCOPE

This document is prepared for design of Foundation for Concentrate Filtration Tanks. Design of concrete structures and foundations has been done in accordance with SBC-304, ACI-318M and API 650.

3 DESIGN LOAD

All Design Loads are indicated in Foundation and Load Plan (Dwg. No. MD-216-3340-EG-ME-GAD-1023). GS DRAWING FOR 3350-REA-001 and 3360-REA-001 attached. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001),Clause 4.2.

4 DESIGN CRITERIA

4.1 Material Specifications

Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Grade slab and pavement: 30 MPa

Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.2 Foundations for Above ground Tanks

Foundation shall be Earth Foundation with a Concrete Ringwall. The design calculations are prepared as per recommendation of standard API 650 - Annex B.

4.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	75

4.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.5m depth below EGL (ie. RL 910.50m). Please refer Table 5 - BH-07 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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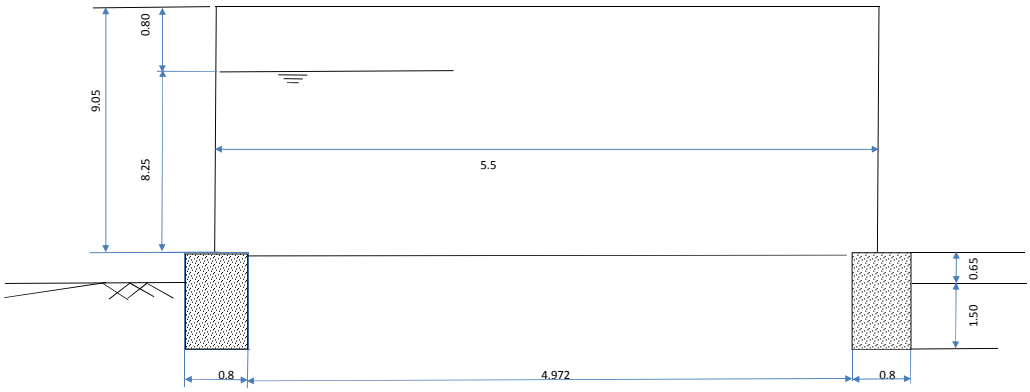
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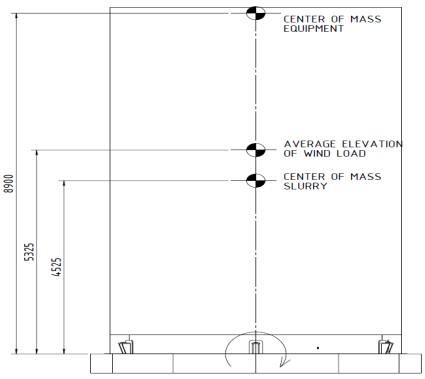
5.0 Design of Foundation for Concentrate Filter Feed Tank (3350-REA-001)

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (Ref: MD-216-3350-EG-ME-PLN-1002)

Tank Loading Data			
	Loading Type		
3350-REA-001	Empyt (W)	191.2	kN
	Weight of platform	60	kN
	Operating (F1)	3194	kN
	Max (Wt)	4250	kN





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Summary of 3350-REA-001 Specification

Diameter of Reactor, D	=	5.5 m	
Height of Reactor, H	=	9.05 m	
Height of Slurry, H_L	=	8.25 m	
Inner Diameter of Concrete Ring Wall, ID	=	4.97 m	
Outer Diameter of Concrete Ring Wall, OD	=	6.57 m	
Height of Concrete Ring Wall, h	=	2.15 m	
Thickness of Concrete Ring Wall, t	=	0.8 m	
P.C.D of Bolts	=	5.77 m	
Top of Foundation	=	0.65 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L	=	16 kN/m^3	(Calculated as Operating wt/Volume)
Unit Weight of Soil, γ_s	=	20 kN/m^3	
Unit Weight of Concrete, γ_c	=	25 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load =	50 kN	As indicated in Load Data
Total Base Shear due to Seismic load =	178 kN	As indicated in Load Data

Seismic Load at actual elevation

Total Base Shear at Center Of Mass Civil Structure =	1	kN	The Base Shears have been calculated from total Seismic Load in proportion of weights.
Total Base Shear at Center Of Mass Equipment =	10	kN	
Total Base Shear at Center Of Mass Slurry =	167	kN	

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)	
Wind Loads	Normal	50.00	266.25	50 * 5.325
Seismic Loads	Normal	178.00	838.00	(given in Load Data Dwg)

Foundation Weight

a) Ring Wall Weight	=	779.73 kN
a) Inside Soil Weight	=	834.87 kN

Sectional Area

Net cross-sectional area for empty condition, Ae	=	14.51 m^2
Full cross-sectional area for operating condition, Ao=		33.92 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze	=	18.74 m^3
Sectional modulus for operating condition,Zo	=	27.87 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 14.54 kN/m

So, Required Min Wall THK.= 0.2631 m = 263.1 mm < 800 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E) - Since Seismic Loads act in both horizontal directions simultaneously, resultant load of both directions is taken in design. Hence, the given seismic load is multiplied by 1.4 for design.

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	3445.20	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	150.72	50.00	266.25
	Operating	LC103 (2.4.1-3)	D + F1 + W	3445.20	50.00	266.25
		LC104 (2.4.1-5)	D + F1+ 0.7E	3445.20	174.44	821.24
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	2067.12	174.44	821.24
	Maximum weight	LC106 (2.4.1-1)	D + F2	4501.20	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{min}= V/A - M/Z kN/m^2	Q_{all}/Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	5059.80	0.00	348.79	348.79	1.29	OK
LC102	1765.32	373.75	65.45	38.63	9.14	OK
LC103	5059.80	373.75	162.57	135.75	3.68	OK
LC104	5059.80	1196.29	192.09	106.23	3.12	OK
LC105	3681.72	1196.29	151.46	65.61	3.95	OK
LC106	6115.80	0.00	180.29	180.29	2.50	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

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Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	5059.80	0.4	2023.92	0.00	-	-
LC102	1765.32	0.4	706.13	50.00	14.1	OK
LC103	5059.80	0.4	2023.92	50.00	40.5	OK
LC104	5059.80	0.4	2023.92	174.44	11.6	OK
LC105	3681.72	0.4	1472.69	174.44	8.4	OK
LC106	6115.80	0.4	2446.32	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment
The resultant Safety Factor should be 1.5 minimum
Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	5059.80	3.286	16626.52	0.00	-	-
LC102	1765.32	3.286	5800.85	373.75	15.5	OK
LC103	5059.80	3.286	16626.52	373.75	44.5	OK
LC104	5059.80	3.286	16626.52	1196.29	13.9	OK
LC105	3681.72	3.286	12098.15	1196.29	10.1	OK
LC106	6115.80	3.286	20096.53	0.00	-	-

Ultimate Strength Design

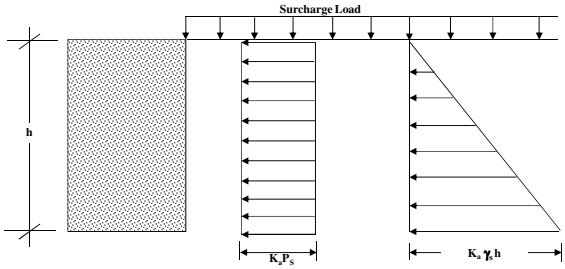
Load Combination (As per SBC-304 , Table 5.3.1)

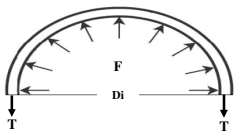
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	4823.28	0.00	23.76	16.33
Short term	Empty	LC102	0.9 D + 1.6*W	226.08	426.00	23.76	16.33
	Operating	LC103	1.2 (D + F1) + 1.6*W	4134.24	426.00	23.76	16.33
		LC104	1.2 (D + F1) + E	4134.24	1173.20	23.76	16.33
		LC105	0.9 (D + F1) + E	3100.68	1173.20	23.76	16.33
	Test	LC106	1.4 (D + F2)	6301.68	0.00	23.76	16.33

Surcharge Load

Load Case	Loads at bottom of Tank level V (kN)	M(kNm)	Surcharge= V/A + M/Z kN/m^2
LC101	4823.28	0.00	203.01
LC102	226.08	426.00	35.60
LC103	4134.24	426.00	200.09
LC104	4134.24	1173.20	245.84
LC105	3100.68	1173.20	202.34
LC106	6301.68	0.00	265.24

Maximum Surcharge Load, Ps = 265.24 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 132.62 KN/m^2

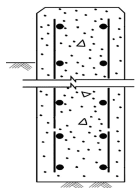
P2 = 1.4 * γ_s * h * K_a = 30.10 KN/m^2 for LC106 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 317.49 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 789.28 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)
Fy = 420 MPa

Required As = T/(ϕ x Fy) = 2088.1 mm^2 (From hoop tension)
Minimum As = 0.0025 x t x h = 4300 mm^2 (As per API 650 , Cl. No. B.4.2.3)
Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2580 mm^2 (As per API 650 , Cl. No. B.4.2.3)
So, Used As = 2 row 8 bar D16 (@ 125mm)
=3217mm2 > Required As OK

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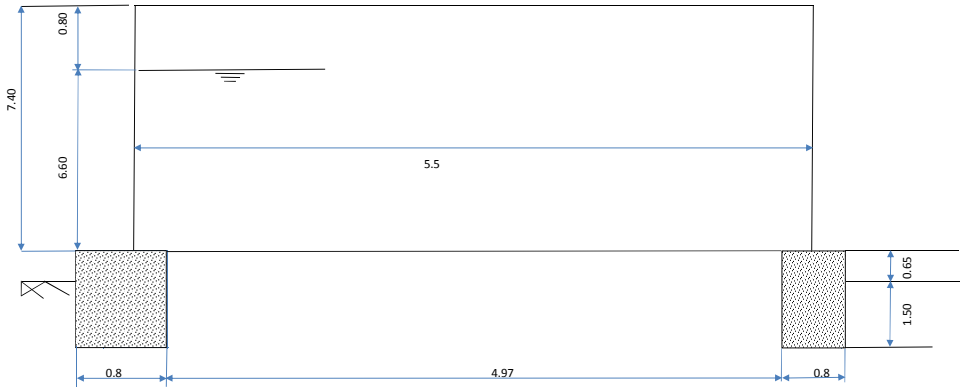
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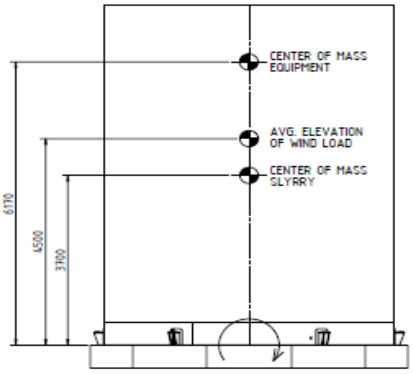
6.0 Design of Foundation for Concentrate Repulp Tank (3360-REA-001)

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (Ref: MD-216-3360-EG-ME-PLN-1001)

Tank Loading Data			
	Loading Type		
3360-REA-001	Empyt (W)	179	kN
	Weight of platform	59	kN
	Operating (F1)	2422	kN
	Max (Wt)	2716	kN





Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023
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Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Summary of Tank Specification

Diameter of Reactor, D	=	5.5 m	
Height of Reactor, H	=	7.4 m	
Height of Slurry, H_L	=	6.6 m	
Inner Diameter of Concrete Ring Wall, ID	=	4.97 m	
Outer Diameter of Concrete Ring Wall, OD	=	6.57 m	
Height of Concrete Ring Wall, h	=	2.15 m	
Thickness of Concrete Ring Wall, t	=	0.8 m	
P.C.D of Bolts	=	5.77 m	
Top of Foundation	=	0.65 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L	=	16 kN/m^3	(Calculated as Operating wt/Volume)
Unit Weight of Soil, γ_s	=	20 kN/m^3	
Unit Weight of Concrete, γ_c	=	25 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load =	41 kN	As indicated in Load Data
Total Base Shear due to Seismic load =	116 kN	As indicated in Load Data

Seismic Load at actual elevation

Total Base Shear at Center Of Mass Civil Structure =	2	kN	} The Base Shears have been calculated from total Seismic Load in proportion of weights.
Total Base Shear at Center Of Mass Equipment =	8	kN	
Total Base Shear at Center Of Mass Slurry =	106	kN	

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)	
Wind Loads	Normal	41.00	184.50	41 * 4.5
Seismic Loads	Normal	116.00	446.00	(given in Load Data Dwg)

Foundation Weight

a) Ring Wall Weight	=	779.46 kN
a) Inside Soil Weight	=	834.20 kN

Sectional Area

Net cross-sectional area for empty condition, Ae =	14.50 m^2
Full cross-sectional area for operating condition, Ao=	33.90 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze	=	18.72 m^3
Sectional modulus for operating condition,Zo	=	27.84 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 13.77 kN/m

So, Required Min Wall THK.= 0.3276 m = 327.6 mm < 800 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E) - Since Seismic Loads act in both horizontal directions simultaneously, resultant load of both directions is taken in design. Hence, the given seismic load is multiplied by 1.4 for design.

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	2660.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	142.80	41.00	184.50
	Operating	LC103 (2.4.1-3)	D + F1 + W	2660.00	41.00	184.50
		LC104 (2.4.1-5)	D + F1+ 0.7E	2660.00	113.68	437.08
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	1596.00	113.68	437.08
	Maximum weight	LC106 (2.4.1-1)	D + F2	2954.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{min}= V/A - M/Z kN/m^2	Q_{all}/Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101 (2.4.1-1)	4273.66	0.00	294.70	294.70	1.53	OK
LC102 (2.4.1-7)	1756.46	272.65	61.60	42.02	9.72	OK
LC103 (2.4.1-3)	4273.66	272.65	135.85	116.27	4.41	OK
LC104 (2.4.1-5)	4273.66	681.49	150.54	101.58	3.98	OK
LC105 (2.4.1-8)	3209.66	681.49	119.15	70.20	5.02	OK
LC106 (2.4.1-1)	4567.66	0.00	134.73	134.73	3.34	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
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Approved	PRI	PRI	PRI	PRI

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	4273.66	0.4	1709.46	0.00	-	-
LC102	1756.46	0.4	702.58	41.00	17.1	OK
LC103	4273.66	0.4	1709.46	41.00	41.7	OK
LC104	4273.66	0.4	1709.46	113.68	15.0	OK
LC105	3209.66	0.4	1283.86	113.68	11.3	OK
LC106	4567.66	0.4	1827.06	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment
The resultant Safety Factor should be 1.5 minimum
Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	4273.66	3.285	14038.98	0.00	-	-
LC102	1756.46	3.285	5769.98	272.65	21.2	OK
LC103	4273.66	3.285	14038.98	272.65	51.5	OK
LC104	4273.66	3.285	14038.98	681.49	20.6	OK
LC105	3209.66	3.285	10543.74	681.49	15.5	OK
LC106	4567.66	3.285	15004.77	0.00	-	-

Ultimate Strength Design

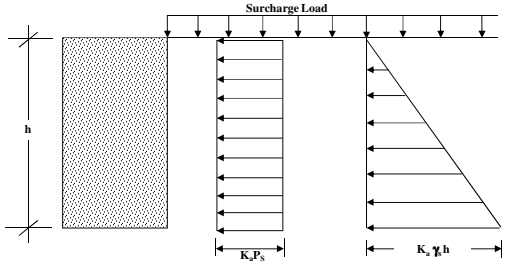
Load Combination (As per SBC-304 , Table 5.3.1)

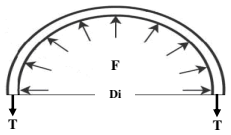
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	3724.00	0.00	23.76	16.33
Short term	Empty	LC102	0.9 D + 1.6*W	214.20	295.20	23.76	16.33
	Operating	LC103	1.2 (D + F1) + 1.6*W	3192.00	295.20	23.76	16.33
		LC104	1.2 (D + F1) + E	3192.00	624.40	23.76	16.33
		LC105	0.9 (D + F1) + E	2394.00	624.40	23.76	16.33
	Test	LC106	1.4 (D + F2)	4135.60	0.00	23.76	16.33

Surcharge Load

Load Case	Loads at bottom of Tank level V (kN)	M(kNm)	Surcharge= V/A + M/Z kN/m^2
LC101	3724.00	0.00	156.75
LC102	214.20	295.20	27.09
LC103	3192.00	295.20	152.43
LC104	3192.00	624.40	172.58
LC105	2394.00	624.40	138.99
LC106	4135.60	0.00	174.07

Maximum Surcharge Load, Ps = 174.07 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

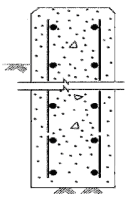
P1 = $K_a * P_s$ = 87.03 KN/m^2

P2 = $1.4 * \gamma_s * h * K_a$ = 30.10 KN/m^2 for LC106 - 1.4 (D + F2)

Maximum Lateral Load
F = P1 * h + 1/2 * P2 * h = 219.48 KN/m

Hoop Tension Load
T= 1/2 * F * ID = 545.41 KN

Hoop Tension Bar
1. Horizontal Reinfocement
ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)
Fy = 420 MPa
Required As = T/(ϕ x Fy) = 1442.9 mm^2 (From hoop tension)
Minimum As = 0.0025 x t x h = 4300 mm^2 (As per API 650 , Cl. No. B.4.2.3)
Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK



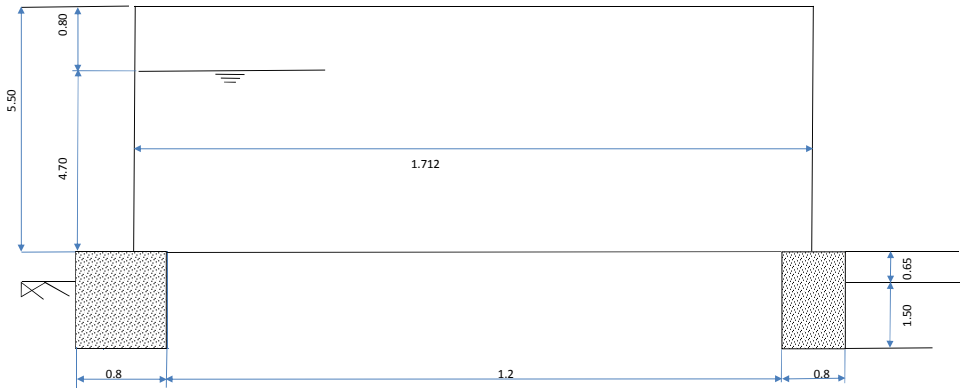
2. Vertical Reinfocement
Minimum As = 0.0015 * b * D = 2580 mm^2 (As per API 650 , Cl. No. B.4.2.3)
So, Used As = 2 row 8 bar D16 (@ 125mm)
=3217mm2 > Required As OK

7.0 Design of Foundation for Concentrate Thickener Overflow Tank (3340-TNK-001)

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (Ref: MD-216-3340-EG-ME-GAD-1023)

Tank Loading Data			
	Loading Type		
3340-TNK-001	Empyt (W)	38.7	kN
	Operating (F1)	250	kN
	Max (Wt) (Operating + Live Load)	540	kN



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023
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Checked	SGC	SGC	SGC	SGC
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Summary of Tank Specification

Diameter of Tank, D	=	1.712 m	
Height of Tank, H	=	5.5 m	
Height of Slurry, H_L	=	4.7 m	
Inner Diameter of Concrete Ring Wall, ID	=	1.2 m	
Outer Diameter of Concrete Ring Wall, OD	=	2.8 m	
Height of Concrete Ring Wall, h	=	2.15 m	
Thickness of Concrete Ring Wall, t	=	0.8 m	
P.C.D of Bolts	=	1.998 m	
Top of Foundation	=	0.65 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L	=	16 kN/m^3	(assumed)
Unit Weight of Soil, γ_s	=	20 kN/m^3	
Unit Weight of Concrete, γ_c	=	25 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load =	8 kN	As indicated in Load Data
Total Base Shear due to Seismic load =	7 kN	Appox calculated

Seismic Load at actual elevation

Total Base Shear at Center Of Mass Civil Structure =		kN	The Base Shears have been calculated from total Seismic Load in proportion of weights.
Total Base Shear at Center Of Mass Equipment =		kN	
Total Base Shear at Center Of Mass Slurry =	7	kN	

(Ref: MD-216-3340-EG-ME-GAD-1024)

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Normal	8.00	23.00
Seismic Loads	Normal	7.00	18.00

Foundation Weight

a) Ring Wall Weight	=	270.18 kN
a) Inside Soil Weight	=	48.63 kN

Sectional Area

Net cross-sectional area for empty condition, Ae =	5.03 m^2
Full cross-sectional area for operating condition, Ao=	6.16 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze =	2.08 m^3
Sectional modulus for operating condition,Zo =	2.16 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2 w / (\gamma_L * H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 7.20 kN/m

So, Required Min Wall THK.= 0.2680 m = 268.0 mm < 800 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E) - Since Seismic Loads act in both horizontal directions simultaneously, resultant load of both directions is taken in design. Hence, the given seismic load is multiplied by 1.4 for design.

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	288.70	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	38.70	8.00	23.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	288.70	8.00	23.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	288.70	6.86	17.64
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	288.70	6.86	17.64
	Maximum weight	LC106 (2.4.1-1)	D + F2	578.70	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Q_{min}= V/A - M/Z kN/m^2	Q_{all}/Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	607.51	0.00	120.86	120.86	3.72	OK
LC102	357.51	40.20	76.71	39.41	7.80	OK
LC103	607.51	40.20	117.31	80.01	5.10	OK
LC104	607.51	32.39	113.69	83.63	5.26	OK
LC105	607.51	32.39	113.69	83.63	5.26	OK
LC106	897.51	0.00	145.76	145.76	3.09	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023
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Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	607.51	0.4	243.00	0.00	-	-
LC102	357.51	0.4	143.00	8.00	17.9	OK
LC103	607.51	0.4	243.00	8.00	30.4	OK
LC104	607.51	0.4	243.00	6.86	35.4	OK
LC105	607.51	0.4	243.00	6.86	35.4	OK
LC106	897.51	0.4	359.00	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment
The resultant Safety Factor should be 1.5 minimum
Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	607.51	1.4	850.51	0.00	-	-
LC102	357.51	1.4	500.51	40.20	12.5	OK
LC103	607.51	1.4	850.51	40.20	21.2	OK
LC104	607.51	1.4	850.51	32.39	26.3	OK
LC105	607.51	1.4	850.51	32.39	26.3	OK
LC106	897.51	1.4	1256.51	0.00	-	-

Ultimate Strength Design

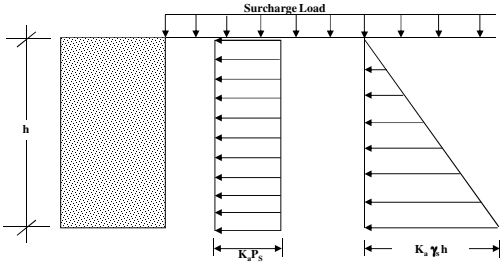
Load Combination (As per SBC-304 , Table 5.3.1)

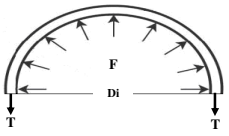
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	404.18	0.00	2.30	0.49
Short term	Empty	LC102	0.9 D + 1.6*W	34.83	36.80	2.30	0.49
	Operating	LC103	1.2 (D + F1) + 1.6*W	346.44	36.80	2.30	0.49
		LC104	1.2 (D + F1) + E	346.44	25.20	2.30	0.49
		LC105	0.9 (D + F1) + E	259.83	25.20	2.30	0.49
	Test	LC106	1.4 (D + F2)	810.18	0.00	2.30	0.49

Surcharge Load

Load Case	Loads at bottom of Tank level V (kN)	M(kNm)	Surcharge= V/A + M/Z kN/m^2
LC101	404.18	0.00	175.58
LC102	34.83	36.80	89.83
LC103	346.44	36.80	225.20
LC104	346.44	25.20	201.65
LC105	259.83	25.20	164.03
LC106	810.18	0.00	351.95

Maximum Surcharge Load, Ps = 351.95 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 175.98 KN/m^2

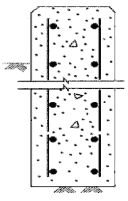
P2 = $1.4 * \gamma_S * h * K_a$ = 30.10 KN/m^2 for LC106 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 410.71 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 246.42 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = $T/(\phi \times Fy)$ = 651.9 mm^2 (From hoop tension)

Minimum As = 0.0025 x t x h = 4300 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2580 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row 8 bar D16 (@ 125mm)

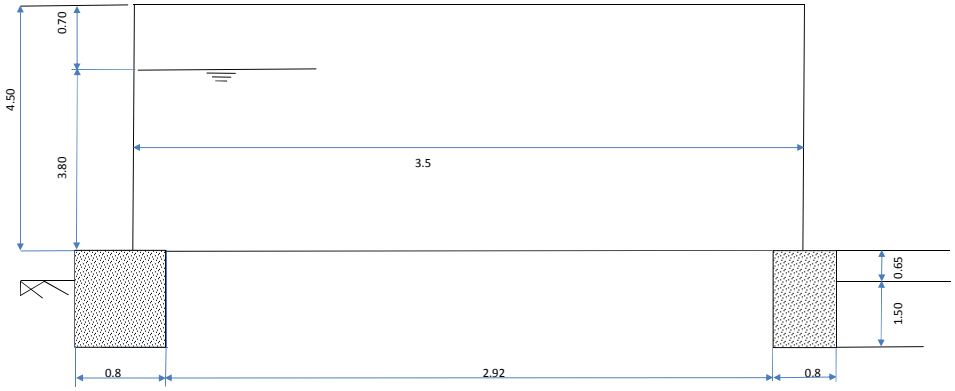
=3217mm2 > Required As OK

8.0 Design of Foundation for Filtrate Tank (3350-TNK-001)

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (Ref: MD-216-3340-EG-ME-GAD-1023)

Tank Loading Data			
	Loading Type		
3350-TNK-001	Empyt (W)	65	kN
	Operating (F1)	392	kN
	Max (Wt) (Operating + Live Load)	917	kN



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
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Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Summary of Tank Specification

Diameter of Tank, D	=	3.5 m	(For Tank Details,
Height of Tank, H	=	4.5 m	Refer dwg no. MD-216-3350-EG-ME-GAD-1027)
Height of Slurry, H_L	=	3.8 m	
Inner Diameter of Concrete Ring Wall, ID	=	2.92 m	
Outer Diameter of Concrete Ring Wall, OD	=	4.52 m	
Height of Concrete Ring Wall, h	=	2.15 m	
Thickness of Concrete Ring Wall, t	=	0.8 m	
P.C.D of Bolts	=	3.72 m	(Appox)
Top of Foundation	=	0.65 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L	=	10 kN/m^3	(Calculated as FULL wt/Volume)
Unit Weight of Soil, γ_s	=	20 kN/m^3	
Unit Weight of Concrete, γ_c	=	25 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load	=	16 kN	As indicated in Load Data
Total Base Shear due to Seismic load	=	121 kN (1.4 * 86)	Appox calculated*

Seismic Load at actual elevation

Total Base Shear at Center Of Mass Civil Structure	=	kN	The Base Shears have been calculated from total Seismic Load in proportion of weights.
Total Base Shear at Center Of Mass Equipment	=	kN	
Total Base Shear at Center Of Mass Slurry	=	121 kN	

* Since Seismic Loads act in both horizontal directions simultaneously, resultant load of both directions is taken in design. Hence, the given seismic load is multiplied by 1.4 for design.

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)	
Wind Loads	Normal	16.00	36.00	16 * 2.25
Seismic Loads	Normal	121.00	272.25	121 * 2.25

Foundation Weight

a) Ring Wall Weight	=	502.53 kN
a) Inside Soil Weight	=	287.95 kN

Sectional Area

Net cross-sectional area for empty condition, Ae	=	9.35 m^2
Full cross-sectional area for operating condition, Ao	=	16.05 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze	=	7.49 m^3
Sectional modulus for operating condition,Zo	=	9.07 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma_c h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma_s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 5.91 kN/m

So, Required Min Wall THK.= 0.7165 m = 716.5 mm < 800 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E) - Since Seismic Loads act in both horizontal directions simultaneously, resultant load of both directions is taken in design. Hence, the given seismic load is multiplied by 1.4 for design.

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	457.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	65.00	16.00	36.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	457.00	16.00	36.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	457.00	84.70	190.58
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	274.20	84.70	190.58
	Maximum weight	LC106 (2.4.1-1)	D + F2	982.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z	Q_{min}= V/A - M/Z	Q_{all}/ Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)	kN/m^2	kN/m^2		
LC101	1247.48	0.00	133.43	133.43	3.37	OK
LC102	855.48	70.40	61.08	45.55	9.80	OK
LC103	1247.48	70.40	85.51	69.98	7.00	OK
LC104	1247.48	372.68	118.85	36.64	5.04	OK
LC105	1064.68	372.68	107.46	25.24	5.57	OK
LC106	1772.48	0.00	110.46	110.46	4.07	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023
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Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
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Approved	PRI	PRI	PRI	PRI

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	1247.48	0.4	498.99	0.00	-	-
LC102	855.48	0.4	342.19	16.00	21.4	OK
LC103	1247.48	0.4	498.99	16.00	31.2	OK
LC104	1247.48	0.4	498.99	84.70	5.9	OK
LC105	1064.68	0.4	425.87	84.70	5.0	OK
LC106	1772.48	0.4	708.99	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment
The resultant Safety Factor should be 1.5 minimum
Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	1247.48	2.26	2819.31	0.00	-	-
LC102	855.48	2.26	1933.39	70.40	27.5	OK
LC103	1247.48	2.26	2819.31	70.40	40.0	OK
LC104	1247.48	2.26	2819.31	372.68	7.6	OK
LC105	1064.68	2.26	2406.19	372.68	6.5	OK
LC105	1772.48	2.26	4005.81	0.00	-	-

Ultimate Strength Design

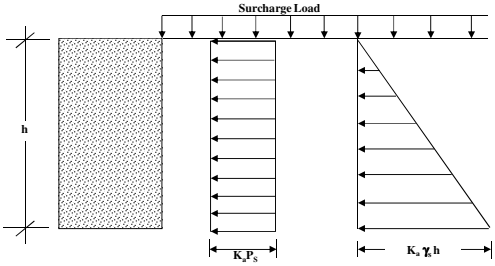
Load Combination (As per SBC-304 , Table 5.3.1)

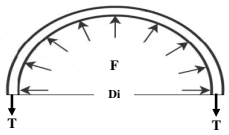
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	639.80	0.00	9.62	4.21
Short term	Empty	LC102	0.9 D + 1.6*W	58.50	57.60	9.62	4.21
	Operating	LC103	1.2 (D + F1) + 1.6*W	548.40	57.60	9.62	4.21
		LC104	1.2 (D + F1) + E	548.40	272.25	9.62	4.21
		LC105	0.9 (D + F1) + E	411.30	272.25	9.62	4.21
	Test	LC106	1.4 (D + F2)	1374.80	0.00	9.62	4.21

Surcharge Load

Load Case	Loads at bottom of Tank level V (kN)	M(kNm)	Surcharge= V/A + M/Z kN/m^2
LC101	639.80	0.00	66.50
LC102	58.50	57.60	19.76
LC103	548.40	57.60	70.68
LC104	548.40	272.25	121.68
LC105	411.30	272.25	107.43
LC106	1374.80	0.00	142.89

Maximum Surcharge Load, Ps = 142.89 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 71.45 KN/m^2

P2 = $1.4 * \gamma_S * h * K_a$ = 30.10 KN/m^2 for LC106 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 185.97 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 271.51 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)
Fy = 420 MPa

Required As = T/(ϕ x Fy) = 718.3 mm^2 (From hoop tension)
Minimum As = 0.0025 x t x h = 4300 mm^2 (As per API 650 , Cl. No. B.4.2.3)
Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2580 mm^2 (As per API 650 , Cl. No. B.4.2.3)
So, Used As = 2 row 8 bar D16 (@ 125mm)
=3217mm2 > Required As OK

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

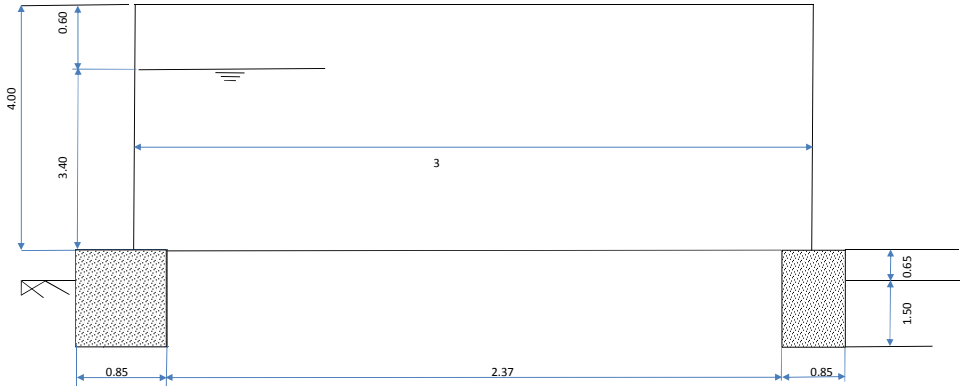
Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023			Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD		
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC		
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI		

9.0 Design of Foundation for Filter Cloth Wash Tank (3350-TNK-002)

(Foundation has been designed as a ring wall on the soil and it follows the principles stated in API 650)

INPUT FOR FOUNDATION: (Ref: MD-216-3340-EG-ME-GAD-1023)

Tank Loading Data			
	Loading Type		
3350-TNK-002	Empyt (W)	50	kN
	Operating (F1)	235	kN
	Max (Wt) (Operating + Live Load)	553	kN



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
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Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Summary of Tank Specification

Diameter of Tank, D	=	3 m	(For Tank Details,
Height of Tank, H	=	4 m	Refer dwg no. MD-216-3350-EG-ME-GAD-1026)
Height of Slurry, H_L	=	3.4 m	
Inner Diameter of Concrete Ring Wall, ID	=	2.37 m	
Outer Diameter of Concrete Ring Wall, OD	=	4.07 m	
Height of Concrete Ring Wall, h	=	2.15 m	
Thickness of Concrete Ring Wall, t	=	0.85 m	
P.C.D of Bolts	=	3.22 m	(Appox)
Top of Foundation	=	0.65 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L	=	10 kN/m^3	(Calculated as FULL wt/Volume)
Unit Weight of Soil, γ_S	=	20 kN/m^3	
Unit Weight of Concrete, γ_C	=	25 kN/m^3	

Wind Load & Seismic Load

Total Base Shear due to wind load	=	12 kN	As indicated in Load Data
Total Base Shear due to Seismic load	=	121 kN (1.4 * 86)	Appox calculated*

Seismic Load at actual elevation

Total Base Shear at Center Of Mass Civil Structure	=	kN	The Base Shears have been calculated from total Seismic Load in proportion of weights.
Total Base Shear at Center Of Mass Equipment	=	kN	
Total Base Shear at Center Of Mass Slurry	=	121 kN	

* Since Seismic Loads act in both horizontal directions simultaneously, resultant load of both directions is taken in design. Hence, the given seismic load is multiplied by 1.4 for design.

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)	
Wind Loads	Normal	12.00	24.00	12 * 2
Seismic Loads	Normal	121.00	242.00	121 * 2

Foundation Weight

a) Ring Wall Weight	=	462.17 kN
a) Inside Soil Weight	=	189.69 kN

Sectional Area

Net cross-sectional area for empty condition, Ae	=	8.60 m^2
Full cross-sectional area for operating condition, Ao	=	13.01 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze	=	5.86 m^3
Sectional modulus for operating condition,Zo	=	6.62 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to reactor tank,minimum thickness of concrete wall required to equalize soil pressures between under reactortank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma c\ h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma c\ h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 5.31 kN/m

So, Required Min Wall THK.= 0.8488 m = 848.8 mm < 850 mm OK

Basic Load Cases

1. Tank Dead Load & Platform load (D)
2. Operational Weight of slurry (F1)
3. Max Weight of slurry (F2)
4. Wind Load (W)
5. Seismic Load (E) - Since Seismic Loads act in both horizontal directions simultaneously, resultant load of both directions is taken in design. Hence, the given seismic load is multiplied by 1.4 for design.

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101 (2.4.1-1)	D + F1	285.00	0	0
Short term	Empty	LC102 (2.4.1-7)	0.6 D + W	50.00	12.00	24.00
	Operating	LC103 (2.4.1-3)	D + F1 + W	285.00	12.00	24.00
		LC104 (2.4.1-5)	D + F1+ 0.7E	285.00	84.70	169.40
		LC105 (2.4.1-8)	0.6 (D + F1) + 0.7E	171.00	84.70	169.40
	Maximum weight	LC106 (2.4.1-1)	D + F2	603.00	0	0

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z	Q_{min}= V/A - M/Z	Q_{all}/Q_{max}	Soil Bearing Check
	V (kN)	M(kNm)	kN/m^2	kN/m^2		
LC101	936.87	0.00	108.96	108.96	4.13	OK
LC102	701.87	49.80	61.47	46.42	9.74	OK
LC103	936.87	49.80	79.53	64.49	7.52	OK
LC104	936.87	351.51	125.12	18.90	4.78	OK
LC105	822.87	351.51	116.36	10.14	5.14	OK
LC106	1254.87	0.00	96.45	96.45	4.67	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	15-09-2020	16-11-2023
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Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	936.87	0.4	374.75	0.00	-	-
LC102	701.87	0.4	280.75	12.00	23.4	OK
LC103	936.87	0.4	374.75	12.00	31.2	OK
LC104	936.87	0.4	374.75	84.70	4.4	OK
LC105	822.87	0.4	329.15	84.70	3.9	OK
LC105	1254.87	0.4	501.95	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment
The resultant Safety Factor should be 1.5 minimum
Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	936.87	2.035	1906.52	0.00	-	-
LC102	701.87	2.035	1428.30	49.80	28.7	OK
LC103	936.87	2.035	1906.52	49.80	38.3	OK
LC104	936.87	2.035	1906.52	351.51	5.4	OK
LC105	822.87	2.035	1674.53	351.51	4.8	OK
LC105	1254.87	2.035	2553.65	0.00	-	-

Ultimate Strength Design

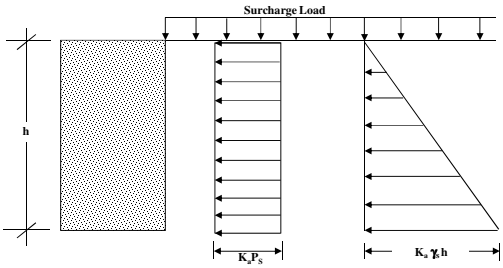
Load Combination (As per SBC-304 , Table 5.3.1)

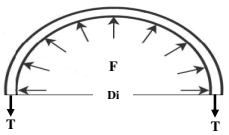
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	399.00	0.00	7.07	2.65
Short term	Empty	LC102	0.9 D + 1.6*W	45.00	38.40	7.07	2.65
	Operating	LC103	1.2 (D + F1) + 1.6*W	342.00	38.40	7.07	2.65
		LC104	1.2 (D + F1) + E	342.00	242.00	7.07	2.65
		LC105	0.9 (D + F1) + E	256.50	242.00	7.07	2.65
	Test	LC106	1.4 (D + F2)	844.20	0.00	7.07	2.65

Surcharge Load

Load Case	Loads at bottom of Tank level V (kN)	M(kNm)	Surcharge= V/A + M/Z kN/m^2
LC101	399.00	0.00	56.45
LC102	45.00	38.40	20.85
LC103	342.00	38.40	62.87
LC104	342.00	242.00	139.68
LC105	256.50	242.00	127.58
LC106	844.20	0.00	119.43

Maximum Surcharge Load, Ps = 139.68 kN/m^2





Where, K_a = coefficient of earth pressure at rest = 0.5

P1 = $K_a * P_s$ = 69.84 KN/m^2

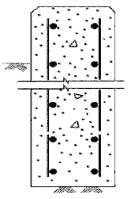
P2 = $1.2 * \gamma_s * h * K_a$ = 25.80 KN/m^2 for LC104 - 1.2 (D + F1) + E

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 177.89 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 210.80 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 557.7 mm^2 (From hoop tension)

Minimum As = 0.0025 x t x h = 4568.75 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D20= 5670.6 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 2741.25 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row 8 bar D16 (@ 125mm)

=3217mm2 > Required As OK

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

REFERENCE

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] API 650 , 12th Edition
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008
[6] Load Plan for Concentrate Thickening Dwg No. MD-216-3340-EG-ME-GAD-1022

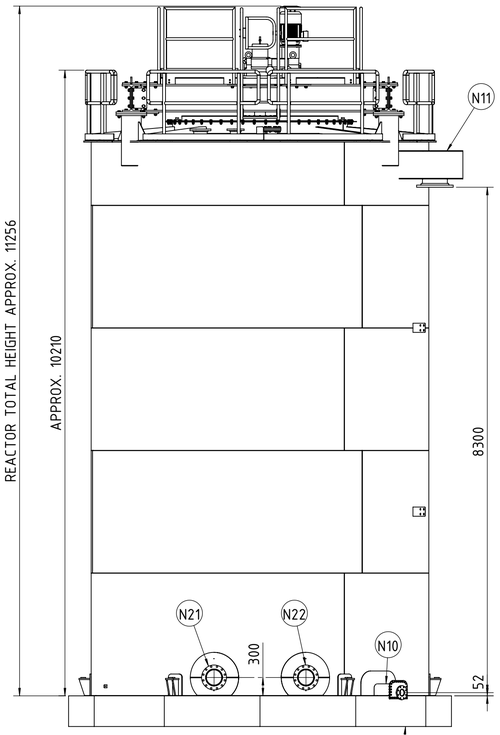
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

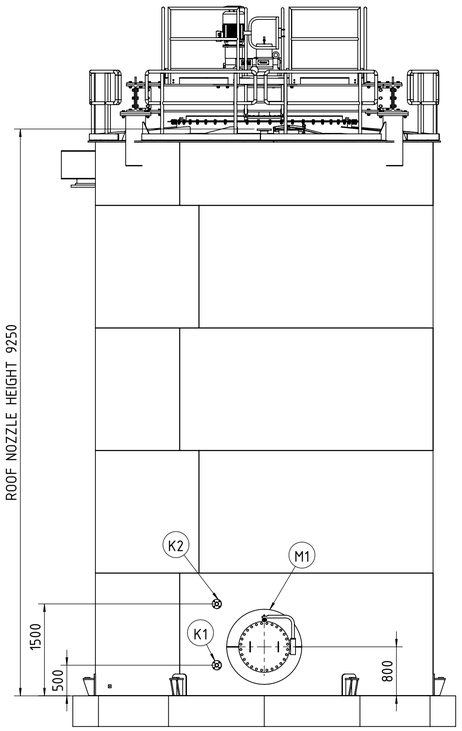
ATTACHMENT 1 GA DRAWING & LOAD DATA OF CONCENTRATE FILTER FEED TANK (3350-REA-001)

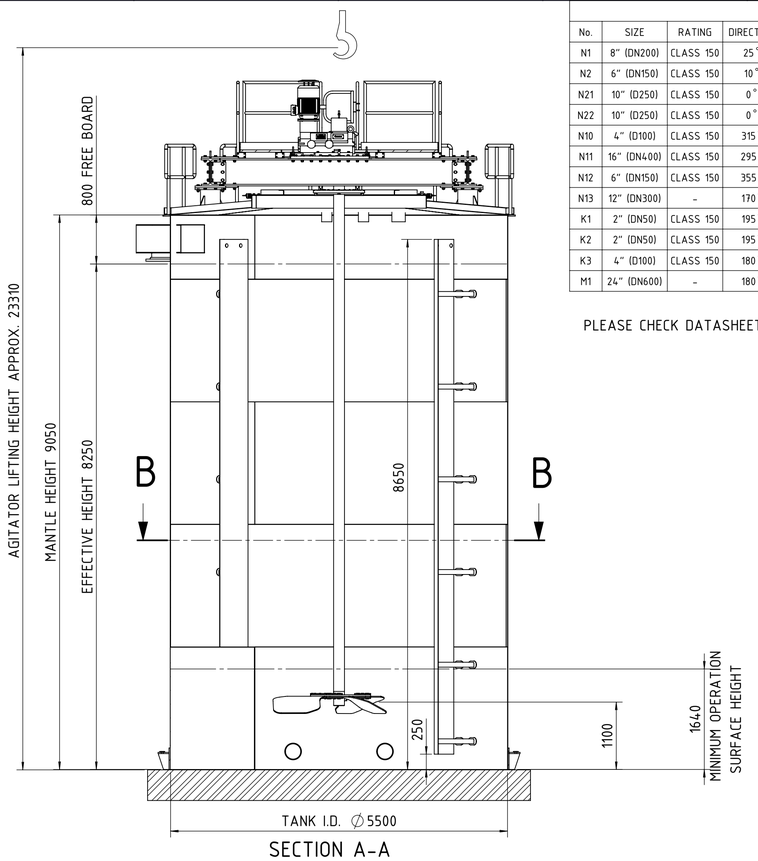
DRAWING NUMBER MD-216-3350-EG-ME-GAD-1052 REV A01



SECTION C-C

CONCRETE BASE NOT INCLUDED IN EQUIPMENT DELIVERY

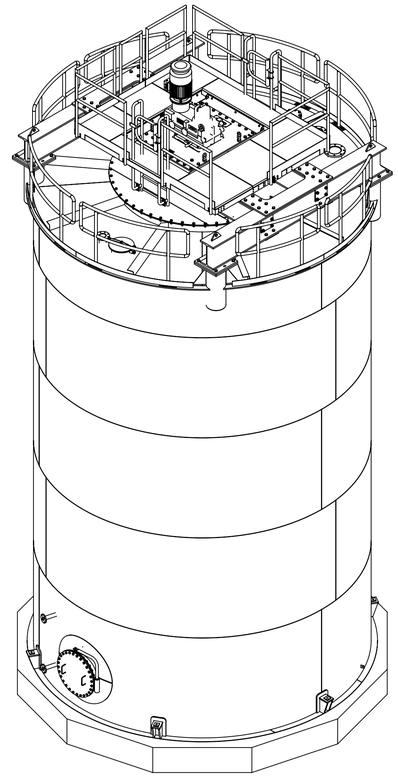


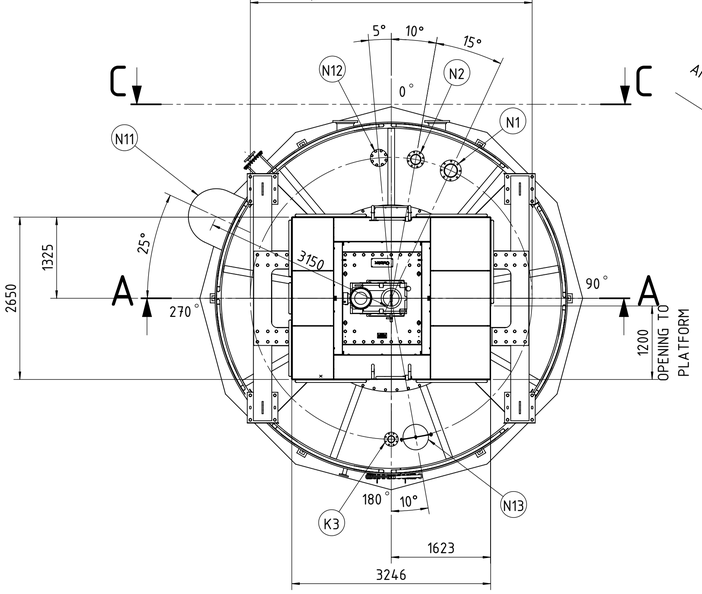


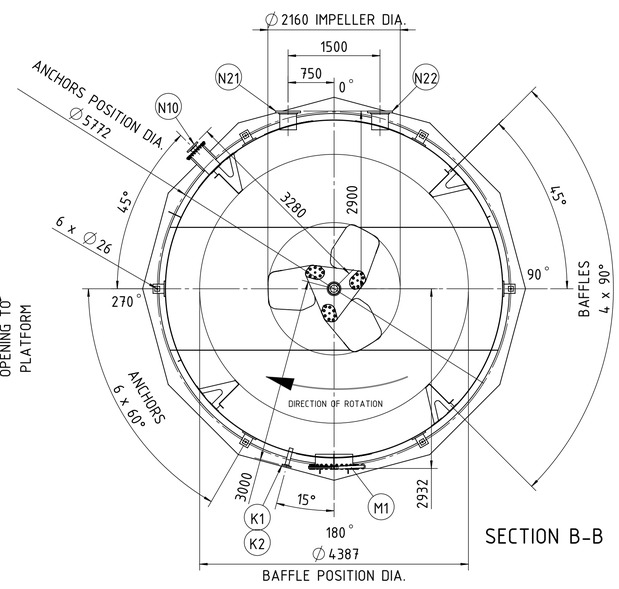
SECTION A-A

NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N1	8" (DN200)	CLASS 150	25°	9250	ASME B16.5	INLET FROM SCREEN	1	
N2	6" (DN150)	CLASS 150	10°	9250	ASME B16.5	CIRCULATION INLET	1	
N21	10" (D250)	CLASS 150	0°	300	ASME B16.5	PUMP OUTLET	1	
N22	10" (D250)	CLASS 150	0°	300	ASME B16.5	PUMP OUTLET	1	
N10	4" (D100)	CLASS 150	315°	78	ASME B16.5	DRAIN	1	
N11	16" (DN400)	CLASS 150	295°	8300	ASME B16.5	EMERGENCY OVERFLOW	1	
N12	6" (DN150)	CLASS 150	355°	9250	ASME B16.5	SPARE	1	
N13	12" (DN300)	-	170°	9250	OUTOTEC	INSPECTION HATCH	1	
K1	2" (DN50)	CLASS 150	195°	500	ASME B16.5	DENSITY MEASUREMENT	1	
K2	2" (DN50)	CLASS 150	195°	1500	ASME B16.5	DENSITY MEASUREMENT	1	
K3	4" (D100)	CLASS 150	180°	9250	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	-	180°	800	OUTOTEC	MANHOLE	1	

PLEASE CHECK DATASHEET MD-216-3350-EG-ME-DAT-1005 FOR PROCESS VALUES







SECTION B-B

CONCENTRATE FILTER FEED TANK	
AGITATOR WEIGHT APPR. (kg)	7896
DEAD WEIGHT (kg)	11631
OPERATIONAL WEIGHT (kg)	345279
MAXIMUM WEIGHT (kg)	453000

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3350-EG-ME-PLN-1002	FOUNDATION AND LOAD PLAN								
MD-216-3350-EG-ME-DAT-1005	EQUIPMENT DATASHEET								
MD-216-3350-EG-PR-PID-1001	P&I DIAGRAM								
MD-216-3350-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	18.12.2019	ISSUED FOR APPROVAL	TM	VH	JR	GR	

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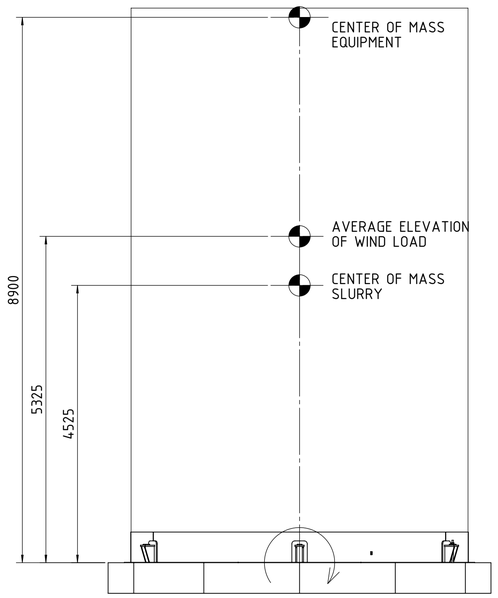
OUTOTEC DRAWING NUMBER 900707 GAU01 SF01 MTL03 33501 DOCUMENT ID OU601247942 REVISION 0

DRAWN	tuoman	17.12.2019
CHECKED	vilhol	18.12.2019
ENGINEER	juhris	18.12.2019
APPROVED	GeoRie	18.12.2019
EMPLOYER		
SCALE	1:50	A1

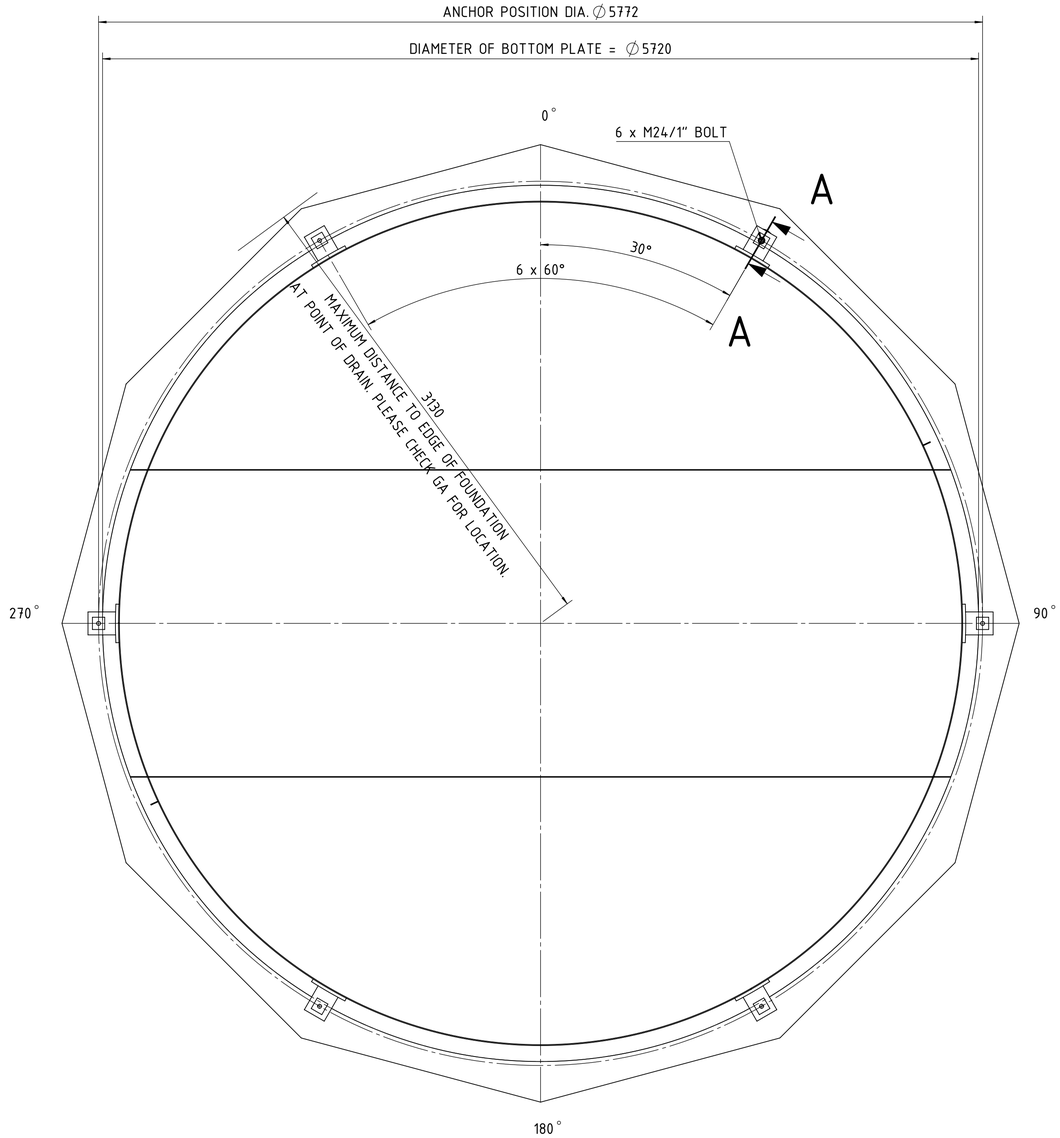
MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CONCENTRATE FILTRATION
3350-REA-001 CONCENTRATE FILTER FEED TANK

DRAWING NUMBER MD-216-3350-EG-ME-GAD-1052 REV A01

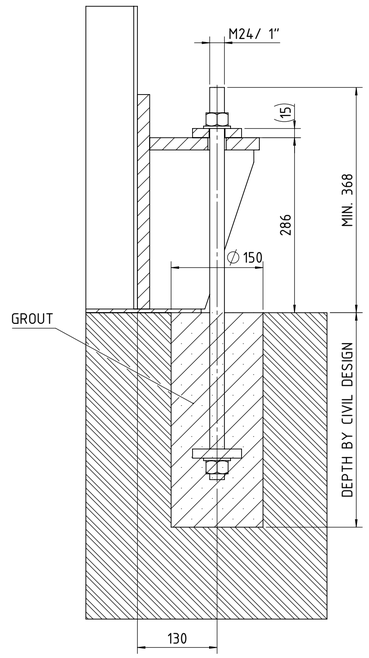
DRAWING NUMBER MD-216-3350-EG-ME-PLN-1002 REV A01



SCALE 1 : 50



LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SECTION A-A
SCALE 1 : 5

CONCENTRATE FILTER FEED TANK	
DEAD WEIGHT OF EQUIPMENT	19500 kg
OPERATIONAL WEIGHT OF SLURRY	325800 kg
MAXIMUM WEIGHT OF SLURRY	433500 kg
EXTERNAL PLATFORM LOAD	5940 kg
NORMAL OPERATING SPEED	50,5 rpm
WIND LOAD	50 kN
SEISMIC LOAD	178 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	838 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3350-EG-ME-GAD-1005	GENERAL ARRANGEMENT DRAWING								
MD-216-3350-EG-ME-DAT-1005	EQUIPMENT DATASHEET								
MD-216-3350-EG-PR-PID-1001	P&I DIAGRAM								
MD-216-3350-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	20.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER 019045-450-M-D-02445-001 REVISION 1

OUTOTEC DRAWING NUMBER 900707 GAU01 SF01 MLD15 33501 DOCUMENT ID OU601261523 REVISION 1

DRAWN	joujaa	20.12.2019
CHECKED	tuoman	20.12.2019
ENGINEER	juhris	20.12.2019
APPROVED	GeoRie	20.12.2019
EMPLOYER		
SCALE	1:20	A1

MANSOURAH-MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
CONCENTRATE FILTRATION
3350-REA-001 CONCENTRATE FILTER FEED TANK,

DRAWING NUMBER MD-216-3350-EG-ME-PLN-1002 REV A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

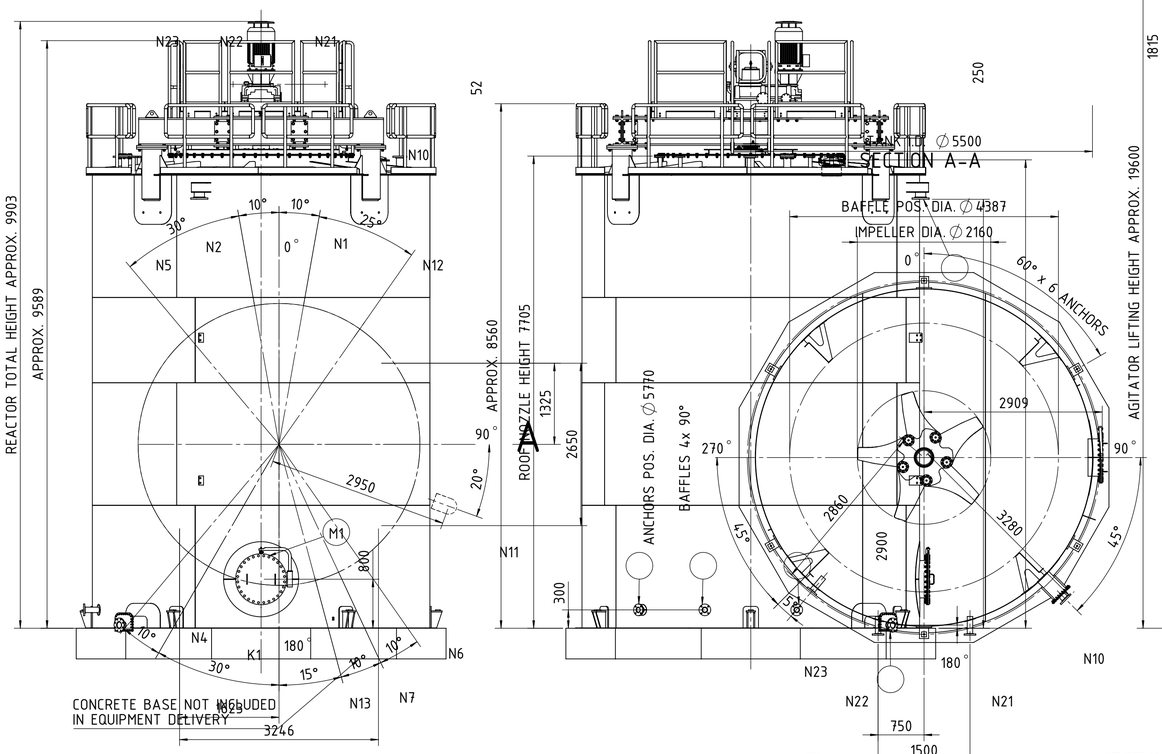
ATTACHMENT 2 GA DRAWING & LOAD DATA OF CONCENTRATE REPLULP TANK (3360-REA-001)

DRAWING NUMBER MD-216-3360-EG-ME-GAD-1001 REV A01

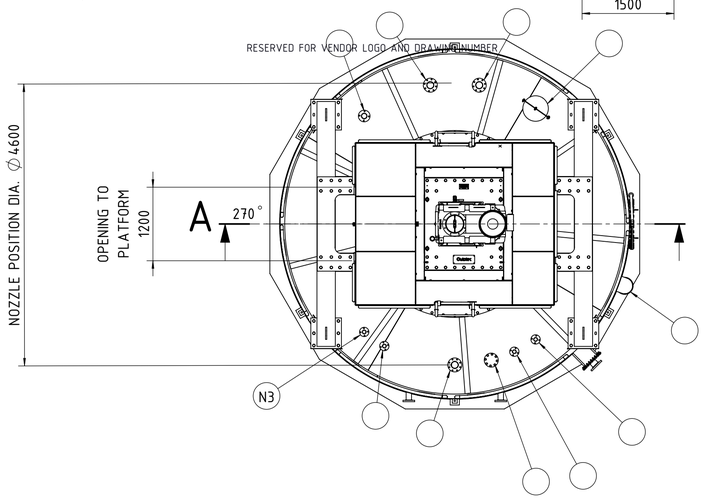
NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N1	4" (DN100)	CLASS 150	10°	7705	ASME B16.5	FILTER CAKE	1	
N2	4" (DN100)	CLASS 150	350°	7705	ASME B16.5	FILTER CAKE	1	
N3	2" (DN50)	CLASS 150	220°	7705	ASME B16.5	DEMIN WATER INLET	1	
N4	2" (DN50)	CLASS 150	210°	7705	ASME B16.5	THICKENER OVERFLOW INLET	1	
N5	3" (DN80)	CLASS 150	320°	7705	ASME B16.5	THICKENER UNDERFLOW	1	
N6	2" (DN50)	CLASS 150	145°	7705	ASME B16.5	SUMP SLURRY INLET	1	
N7	2" (DN50)	CLASS 150	155°	7705	ASME B16.5	DENSITY PUMP INLET	1	
N10	4" (DN100)	-	135°	52	OUTOTEC	DRAIN	1	
N11	6" (DN150)	CLASS 150	110°	7000	ASME B16.5	EMERGENCY OVERFLOW	1	
N12	12" (DN300)	-	35°	7643	OUTOTEC	INSPECTION HATCH	1	
N13	4" (DN100)	CLASS 150	165°	7705	ASME B16.5	SPARE	1	WITH BLIND FLANGE
N21	3" (DN80)	CLASS 150	180°	300	ASME B16.5	PUMP OUTLET	1	
N22	3" (DN80)	CLASS 150	180°	300	ASME B16.5	PUMP OUTLET	1	
N23	3" (DN80)	CLASS 150	220°	300	ASME B16.5	DENSITY PUMP OUTLET	1	
K1	4" (DN100)	CLASS 150	180°	7705	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	-	90°	800	OUTOTEC	MANHOLE	1	

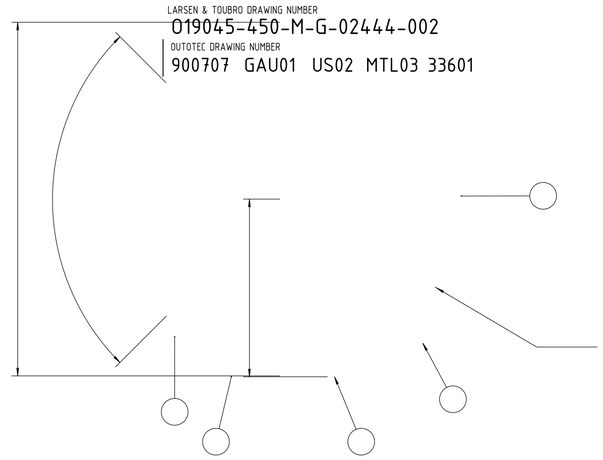
ON HOLD

PLEASE CHECK DATASHEET MD-216-3360-EG-ME-DAT-1001 FOR PROCESS VALUES

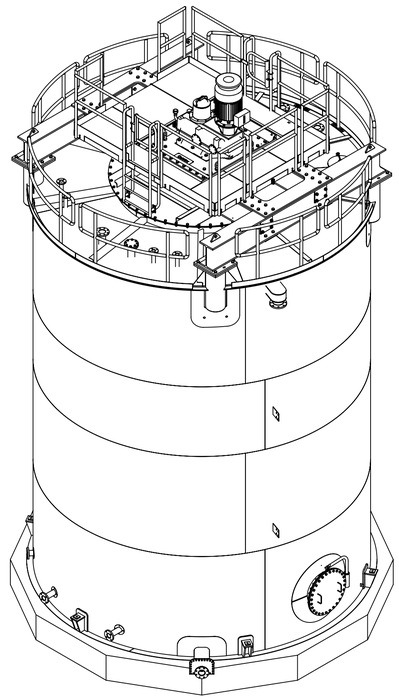


SECTION B-B





LARSEN & TOUBRO DRAWING NUMBER
019045-450-M-G-02444-002
OUTOTEC DRAWING NUMBER
900707 GAU01 US02 MTL03 33601



CONCENTRATE REPULP TANK	
AGITATOR WEIGHT APPR. (kg)	3600
DEAD WEIGHT (kg)	18300
OPERATIONAL WEIGHT (kg)	265300
MAXIMUM WEIGHT (kg)	295200

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3360-EG-ME-PLN-1001	FOUNDATION AND LOAD PLAN								
MD-216-3360-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3360-EG-PR-PID-1010	P&I DIAGRAM								
MD-216-3360-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	20.12.2019	ISSUED FOR APPROVAL	MH	TM	JR	GR	

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DRAWN	mikhov	19.12.2019	
CHECKED	tuoman	20.12.2019	
ENGINEER	juhris	20.12.2019	
APPROVED	GeoRie	20.12.2019	
EMPLOYER			
SCALE	1:50	A1	

REVISION 0

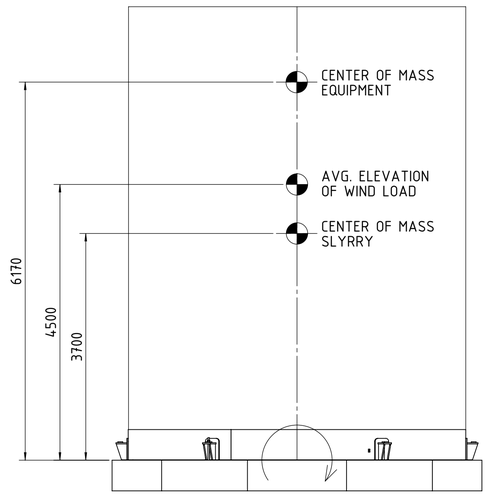
DOCUMENT ID OU601238629 REVISION 0

MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CONCENTRATE STORAGE AND REPULP
3360-REA-001 CONCENTRATE REPULP TANK

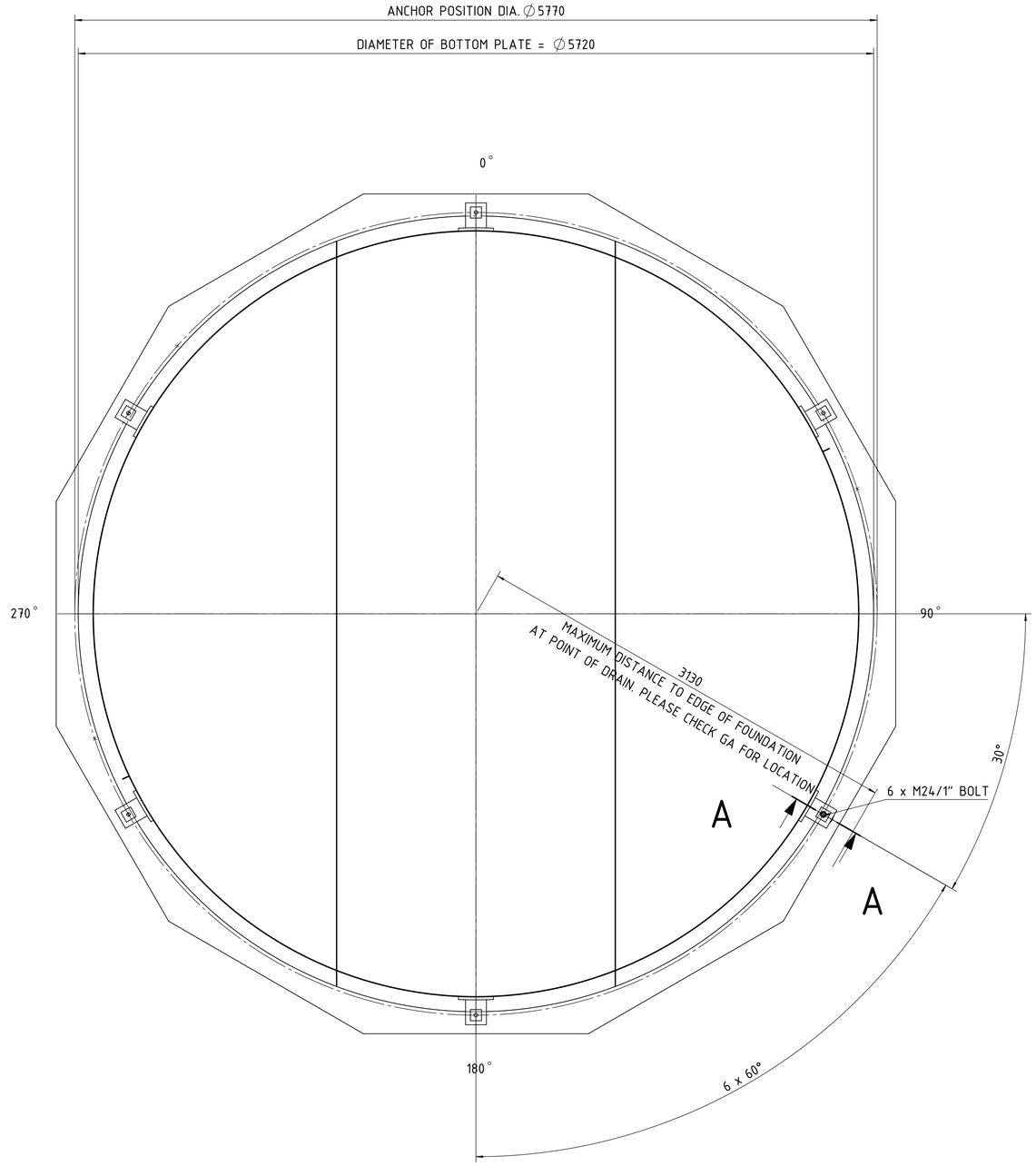
DRAWING NUMBER MD-216-3360-EG-ME-GAD-1001 REV A01

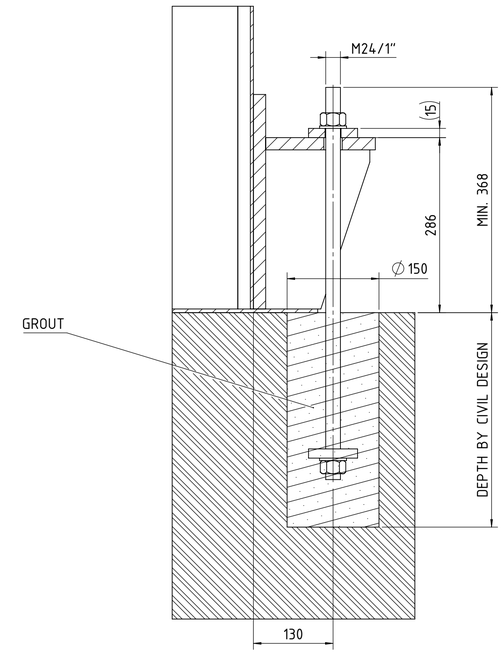
DRAWING NUMBER: MD-216-3360-EG-ME-PLN-1001
REV: A01

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 50





SECTION A-A
SCALE 1 : 5

CONCENTRATE REPULP TANK	
DEAD WEIGHT OF EQUIPMENT	18200 kg
OPERATIONAL WEIGHT OF SLURRY	247000 kg
MAXIMUM WEIGHT OF SLURRY	277000 kg
EXTERNAL PLATFORM LOAD	5940 kg
NORMAL OPERATING SPEED	50,9 rpm
WIND LOAD	41 kN
SEISMIC LOAD	116kN
SEISMIC TORQUE AT FOUNDATION LEVEL	446 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3360-EG-ME-GAD-1001	GENERAL ARRANGEMENT DRAWING								
MD-216-3360-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3360-EG-PR-PID-1010	P&I DIAGRAM								
MD-216-3360-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	20.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-02445-002 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707 GAU01 US02 MLD15 33601 — DOCUMENT ID: OU601262588 — REVISION: 1

DRAWN	joujaa	20.12.2019
CHECKED	tuoman	20.12.2019
ENGINEER	juhris	20.12.2019
APPROVED	GeoRie	20.12.2019
EMPLOYER		
SCALE	1:20	A1

MANSOURAH-MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
CONCENTRATE STORAGE AND REPULP
3360-REA-001 CONCENTRATE REPULP TANK

DRAWING NUMBER: MD-216-3360-EG-ME-PLN-1001 — REV: A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

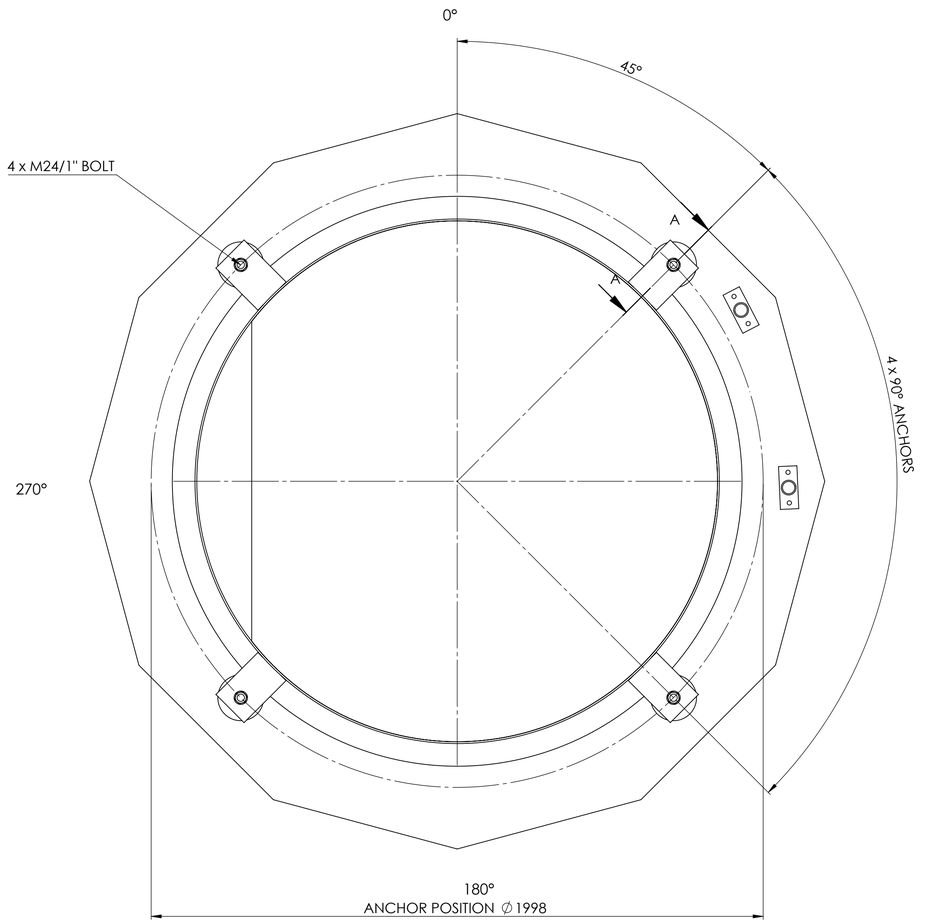
Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023			Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI	

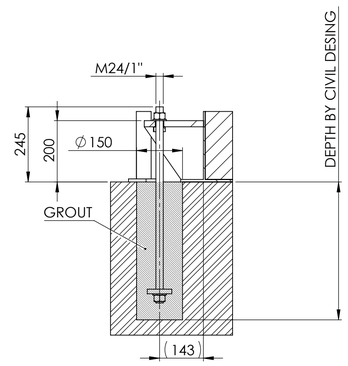
ATTACHMENT 3 LOAD DATA DRAWING OF CONCENTRATE THICKENER OVERFLOW TANK (3340-TNK-001)

DRAWING NUMBER MD-216-3340-EG-ME-GAD-1024
REV A01

FOOTING LOAD TABLE
3340-TNK-001 FLOTATION CONCENTRATE THICKENER OVERFLOW TANK

LOAD	DESCRIPTION	VERTICAL (kN)	HORIZONTAL (kN)	MOMENT (kNm)
1	TANK DEAD LOAD	55	-	-
2	NORMAL OPERATING FLUID LOAD SG=1,0	104	-	-
3	MAX FLUID LOAD SG=1,0	112	-	-
4	SEISMIC LOAD	-	7	18
5	WIND LOAD	-	8	23





SECTION A-A

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

VALUES ARE PRESENTED WITHOUT ANY FACTORS

LOADING NOTES:
1. ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON GROUND OR CONCRETE SLAB)

FOUNDATION AND SUPPORT TOLERANCES:
1. THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2. THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL

CONCRETE NOTES: (WHERE TANKS ARE INSTALLED ON CONCRETE)
1. CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2. FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWN BOLT EDGE DISTANCE TO BE DETERMINED BY CIVIL ENGINEERING

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3340-EG-ME-GAD-1013	OUTLINE DRAWING								
MD-216-3340-EG-PR-PID-1003	P&I DIAGRAM								
MD-216-3340-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	05.02.2020	ISSUED FOR APPROVAL	JR	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-G-02446-034 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 GAU01 ST01 MLD15 33402 — DOCUMENT ID: OU601447945 — REVISION: 0

DRAWN	juhris	05.02.2020
CHECKED	tuoman	05.02.2020
ENGINEER	juhris	05.02.2020
APPROVED	GeoRie	05.02.2020
EMPLOYER		
SCALE	1:10	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
FLOTATION
3340-TNK-001 FLOTATION CONCENTRATE THICKENER OVERFLOW TANK

DRAWING NUMBER: MD-216-3340-EG-ME-GAD-1024 — REV: A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	17-02-2020	17-08-2020	14-09-2020	16-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 13-08-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 4 LOAD TABLE FOR CONCENTRATE THICKENING TANKS

DRAWING NUMBER MD-216-3340-EG-ME-GAD-1023 REV A01

LOADING POINT		P1				P2*				P3				P4				P5				P6				P7				P8				P9				P10*				P11*				P12				P13				P14			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT	-	10							-	2			-	2			-	2			-	10			-	12			-	5			-	5							-	4			-	4			-	40		
2	DEAD LOAD (SECONDARY EQUIPMENT)																																																			
3	OPERATION LOAD 1. FILLING																																																	-	250		
4	OPERATION LOAD 2. PRESSURE																																																		
5	OPERATION LOAD 3. TEMPERATURE																																																		
6	OPERATION LOAD 4. DYNAMIC	-	15							-	4			-	4			-	4			-	15			-	18			-	8			-	8							-	6			-	6						
7	OPERATION LOAD 5. OTHERS																																																		
8	TEST LOAD FILLING / EMERGENCY																																																		
9	LIVE LOAD																																																-	290		
10	WIND ±X - DIRECTION			-	-						-	1	-		-	1	-		-	-	-			1	-		-	1	-		-	1	-							-	1	-		-	1	-		-	10	-
11	WIND ±Y - DIRECTION			-	-						-	-	1.5		-	-	1.5		-	-	-		-	-	1		-	-	1		-	-	1							-	-	1		-	-	1		-	-	10
12	EARTHQUAKE ±X - DIRECTION			0.5	0.5						-	-	-		-	-	-		-	-	-			0.5	0.5		-	0.5	0.5		-	-	-							-	-	-		-	-	-		-	-	-
13	EARTHQUAKE ±Y - DIRECTION			0.5	0.5						-	-	-		-	-	-		-	-	-			0.5	0.5		-	0.5	0.5		-	-	-							-	-	-		-	-	-		-	-	-
	EQUIPMENT - NO.	3360-PMP-101				3360-REA-001				3360-PMP-001				3360-PMP-002				3360-PMP-003				3340-PMP-501				3340-PMP-101				3340-PMP-002				3340-PMP-001				3340-THK-001				3340-THK-001				3340-PMP-004				3340-PMP-003				3340-TNK-001			
	DESCRIPTION	CONC. REPULP AREA SUMP PUMP				CONC. REPULP TANK				CONC. REPULP PUMP				CONC. REPULP PUMP				REPULP DENSITY MEASUREMENT PUMP				FLOT. CONC. SAMPLE TRANSFER PUMP				FLOT. CONC. THICK. AREA SUMP PUMP				FLOT. CONC. THICK. UNDERFLOW PUMP				FLOT. CONC. THICK. UNDERFLOW PUMP				FLOTATION CONCEN- TRATE THICKENER				FLOTATION CONCEN- TRATE THICKENER				FLOT. CONC. THICK. OVERFLOW PUMP				FLOT. CONC. THICK. OVERFLOW PUMP				FLOT. CONC. THICK. OVERFLOW TANK			

LOADING POINT		P15*				P16				P17				P18				P19				P20				P21				P22				P23				P24				P25				P26				P27				P28			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT					-	12			-	14			-	14			-	20			-	20			-	50			-	3			-	3			-	3			-	3			-	65			-	40			-	3		
2	DEAD LOAD (SECONDARY EQUIPMENT)																																																				
3	OPERATION LOAD 1. FILLING																									-	235																					-	392			-	65			-	6		
4	OPERATION LOAD 2. PRESSURE																																																			
5	OPERATION LOAD 3. TEMPERATURE																																																			
6	OPERATION LOAD 4. DYNAMIC					-	18			-	22			-	22			-	30			-	30							-	5			-	5			-	5			-	5							-	8		
7	OPERATION LOAD 5. OTHERS																																																			
8	TEST LOAD FILLING / EMERGENCY																																																			
9	LIVE LOAD																									-	318																					-	525			-	105						
10	WIND ±X - DIRECTION					-	-	1	-	-	-	1	-	-	-	1	-	-	-	10	-	-	-	10	-	-	-	12	-	-	-	1	-	-	-	1	-	-	-	1	-	-	-	1	-	-	-	16	-	-	-	27	-	-	-	6	-
11	WIND ±Y - DIRECTION					-	-	1	-	-	-	1	-	-	-	1	-	-	-	10	-	-	-	10	-	-	-	12	-	-	-	1	-	-	-	1	-	-	-	1	-	-	-	1	-	-	-	16	-	-	-	27	-	-	-	6	-
12	EARTHQUAKE ±X - DIRECTION					-	0.6	0.6		-	0.7	0.7		-	0.7	0.7		-	-	-		-	-	-		-	86	86		-	-	-		-	-	-		-	-	-		-	-	-		-	86	86		-	2	2		-	-	-	
13	EARTHQUAKE ±Y - DIRECTION					-	0.6	0.6		-	0.7	0.7		-	0.7	0.7		-	-	-		-	-	-		-	86	86		-	-	-		-	-	-		-	-	-		-	-	-		-	86	86		-	2	2		-	-	-	
	EQUIPMENT - NO.	3350-REA-001				3350-PMP-102				3350-PMP-005				3350-PMP-006				3350-PMP-002				3350-PMP-001				3350-TNK-002				3350-PMP-004				3350-PMP-003				3350-PMP-008				3350-PMP-007				3350-TNK-001				3340-TNK-003				3340-SMP-002			
	DESCRIPTION	CONCENTRATE FILTER FEED TANK				FILTER FEED TANK SUMP PUMP				FILTER CLOTH WASH PUMP				FILTER CLOTH WASH PUMP				CONC. FILTER FEED PUMP				CONC. FILTER FEED PUMP				FILTER CLOTH WASH TANK				FILTRATE PUMP				FILTRATE PUMP				FILTER CORE FLUSH PUMP				FILTER CORE FLUSH PUMP				FILTRATE TANK				THICKENER FEED TANK				THICKENER FEED SECONDARY SAMPLER			

LOADING POINT		P29				P30				P31				P32				P33				P34				P35.1/2				P36.1/2				P37				P38.1				P38.2			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT	-	15			-	15			-	3			-	5			-	5			-	5			-	20			-	15			-	25			-	10			-	8		
2	DEAD LOAD (SECONDARY EQUIPMENT)																																												
3	OPERATION LOAD 1. FILLING									-	6															-	30			-	20			-	70										
4	OPERATION LOAD 2. PRESSURE																																											
5	OPERATION LOAD 3. TEMPERATURE																																											
6	OPERATION LOAD 4. DYNAMIC	-	25			-	25			-	8							-	8			-	8			-	60			-	45							-	1			-	1		
7	OPERATION LOAD 5. OTHERS																																											
8	TEST LOAD FILLING / EMERGENCY																																											
9	LIVE LOAD																																	-	95										
10	WIND ±X - DIRECTION		-	1	-		-	1	-		-	-	-		-	3	-		-	0.7	-		-	0.7	-		-	0.5	1		-	0.5	1		-	10	-		-	2	-		-	2	-	
11	WIND ±Y - DIRECTION		-	-	2		-	-	2		-	-	-		-	-	3		-	-	0.7		-	-	0.7		-	1	0.5		-	1	0.5		-	-	10		-	-	4		-	-	4
12	EARTHQUAKE ±X - DIRECTION		-	0.75	0.75		-	0.75	0.75		-	-	-		-	-	-		-	-	-		-	-	-		-	1.5	1.5		-	1	1		-	2.75	-		-	-	-		-	-	-
13	EARTHQUAKE ±Y - DIRECTION		-	0.75	0.75		-	0.75	0.75		-	-	-		-	-	-		-	-	-		-	-	-		-	1.5	1.5		-	1	1		-	2.75	-		-	-	-		-	-	-
	EQUIPMENT - NO.	3360-FDR-002				3360-FDR-003				3360-SMP-001				3360-CHT-005				3360-SCB-001				3360-FAN-001				3340-SMP-001				3340-SMP-001				3340-TNK-002				3350-SCR-001				3350-SCR-001			
	DESCRIPTION	CONCENTRATE SCREW PULPER				CONCENTRATE SCREW PULPER				CONC. REPULP SAMPLER				CONC. RECLAM CONVEYOR DIS- CHARGE CHUTE				CONC. REPULP DEMISTER				CONC. REPULP OFFGAS FAN				FLOT. CONC. THI- CKENER FEED PR- IMARY SAMPLER				FLOT. CONC. THI- CKENER FEED PR- IMARY SAMPLER				FLOT. CONC. SAMPLER FEED TANK				CONC. THICKENER FEED TRASH SCREEN				CONC. THICKENER FEED TRASH SCREEN			

* :-
FOR,
P2 :REFER DWG. MD-216-3360-EG-ME-PLN-1001 FOR LOAD DETAILS
P10, P11 : REFER DWG. MD-216-3340-EG-ME-GAD-1002 FOR LOAD DETAILS
P15 :REFER DWG. MD-216-3350-EG-ME-PLN-1002 FOR LOAD DETAILS

0 1 10m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

GENERAL NOTES, LEGEND

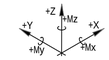
01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS



06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM- GRATING 5 kN/m² GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

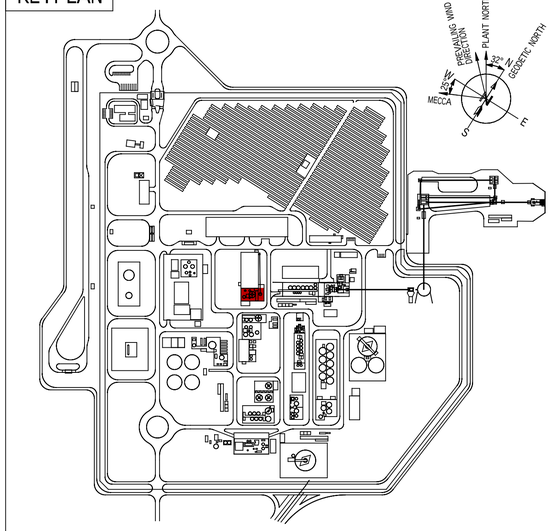
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



REV	DOCUMENT NUMBER	REFERENCE DOCUMENTS	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	MD-216-3340-EG-ME-GAD-1003	PLAN AT EL. +921.750							
	MD-216-3000-EG-GM-LST-1010	HOLD LIST							
	MD-216-3340-EG-ME-GAD-1022	PLAN AT EL. +918.750							
A01	MD-216-3340-EG-ME-GAD-1021	PLAN AT EL. +913.000	30-01-2020	ISSUED FOR APPROVAL	BS	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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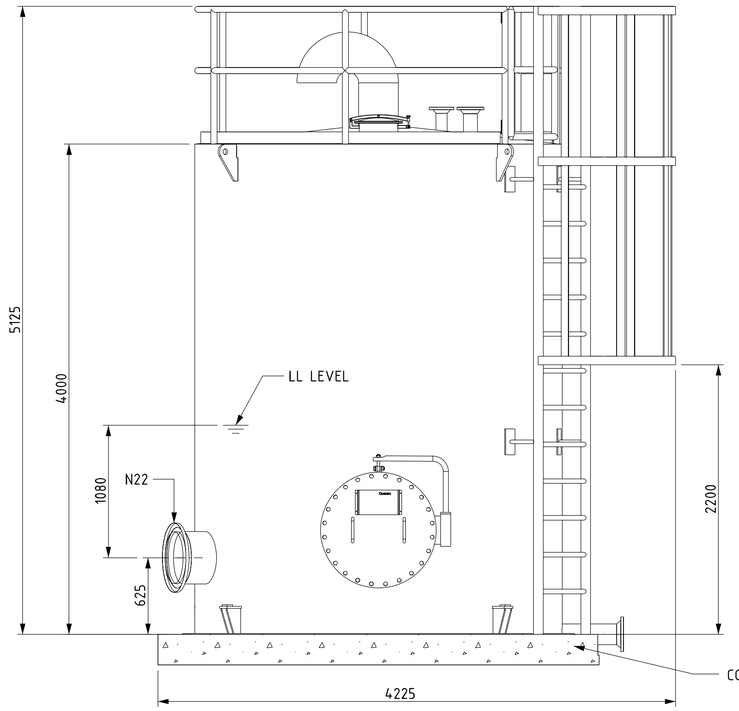
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OUTOTEC DRAWING NUMBER	900707 _GAU01 _ST01 _MLD14 _33404	DOCUMENT ID OU100670406 / REVISION 0

DRAWN	BS	30-01-2020
CHECKED	TV	30-01-2020
ENGINEER	SC	30-01-2020
APPROVED	RL	30-01-2020
EMPLOYER		
SCALE	NA	A1

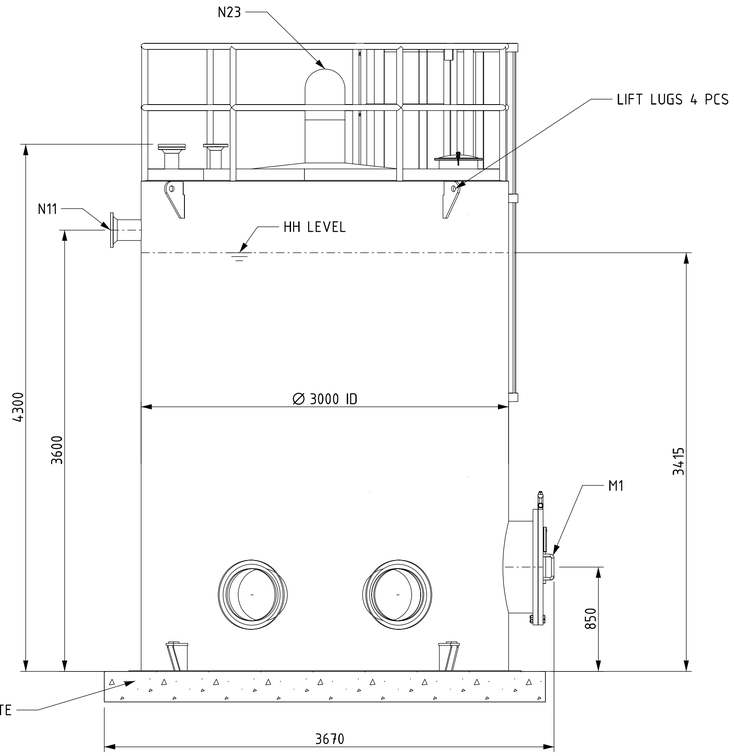
MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
CONCENTRATE THICKENING
LOAD TABLE

DRAWING NUMBER MD-216-3340-EG-ME-GAD-1023 REV A01

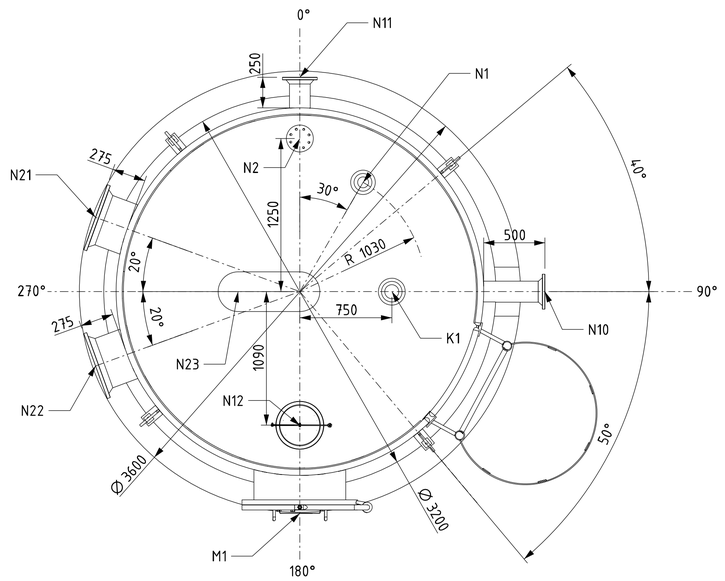
DRAWING NUMBER: MD-216-3350-EG-ME-GAD-1026 | REV: A01



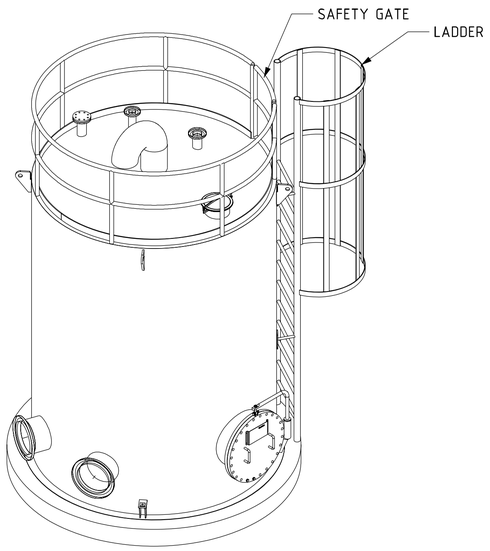
FRONT VIEW



LEFT VIEW



TOP VIEW



ISOMETRIC VIEW 1:40

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3350-TNK-002
DESCRIPTION	FILTER CLOTH WASH TANK
DESIGN CODE	API 650
DESIGN TEMPERATURE	+5-45°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	0,977 kPa/ SBC 301,2007
CORROSION ALLOWANCE: SHELL INSIDE	2 mm
CORROSION ALLOWANCE: BOTTOM	2 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	5125mmx4255mmx3670mm
SHELL HEIGHT	4000
CAPACITY: TOTAL VOLUME	28m³
CAPACITY: TOTAL LIQUID VOLUME	12m³
SEISMIC DESIGN /CODE	MD-216-1000-EG-GE-SPC-1001 /SBC 301, 2007

LOAD DATA

TOTAL WEIGHT (FULL)	32900 kg
TANK WEIGHT	4900 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284/ EN10025-2 S235JR
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A284
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N1	3"	150	30°	250	-	4300	ASME B16.5	INLET, FILTERED WATER
N2	4"	150	0°	250	-	4300	ASME B16.5	SPARE
N21	16"	150	290°	275	-	625	ASME B16.5	OUTLET TO PUMP
N22	16"	150	250°	275	-	625	ASME B16.5	OUTLET TO PUMP
N23	12"	150	270°	-	-	4500	ASME B16.5	VENT
N10	6"	150	90°	500	-	0	ASME B16.5	DRAIN
N11	6"	150	0°	250	-	3600	ASME B16.5	EMERGENCY OVERFLOW
N12	12"	150	180°	-	-	4170	ASME B16.5	INSPECTION HOLE
M1	Ø30"	150	180°	-	-	850	ASME B16.5	MANHOLE
K1	4"	150	90°	-	-	4300	ASME B16.5	LEVEL INDICATOR

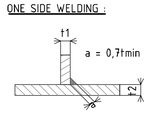
NOTE:
1. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
2. ALL ELEVATION MEASUREMENTS ARE MEASURED FROM BOTTOM SIDE OF BASE PLATE.
3. SHELL NOZZLE ELEVATIONS ARE FROM BOTTOM FROM BOTTOM OF BASE PLATE TO CENTERLINE OF NOZZLE.
4. SHELL NOZZLE PROJECTIONS ARE MEASURED FROM OUTSIDE OF SHELL TO FLANGE FACE.
5. ALL NOZZLE LOCATION, ELEVATION AND ORIENTATION ARE INDICATIVE ONLY AND WILL BE FINALIZED DURING DETAIL DESIGN PHASE.

WELDING GENERAL NOTES

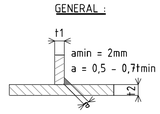
ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING :



GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	31-01-20	ISSUED FOR APPROVAL	AV	TV	SC	RL	

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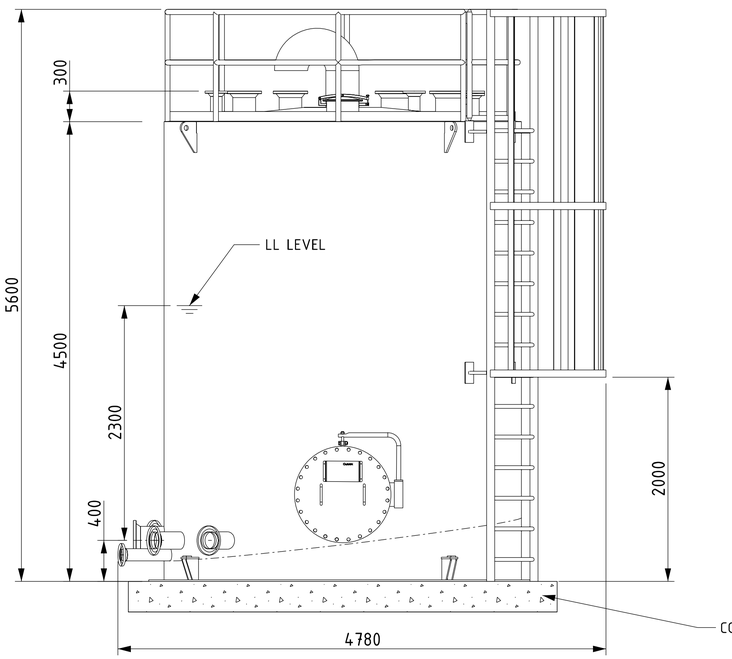
OUTOTEC DRAWING NUMBER: 900707_GAU01_SF01_MTA01_33515 | DOCUMENT ID: OU100661832 | REVISION: 0

DRAWN	AliVai	31-01-20
CHECKED	TatVie	31-01-20
ENGINEER	SouCha	31-01-20
APPROVED	RicLeh	31-01-20
EMPLOYER		
SCALE	1:25	A1

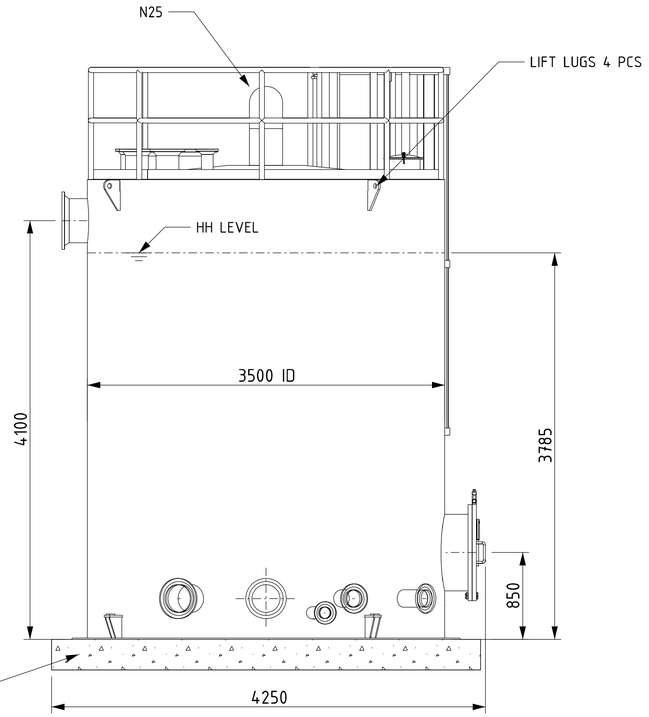
MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
CONCENTRATE FILTRATION
3350-TNK-002 FILTER CLOTH WASH TANK

DRAWING NUMBER: MD-216-3350-EG-ME-GAD-1026 | REV: A01

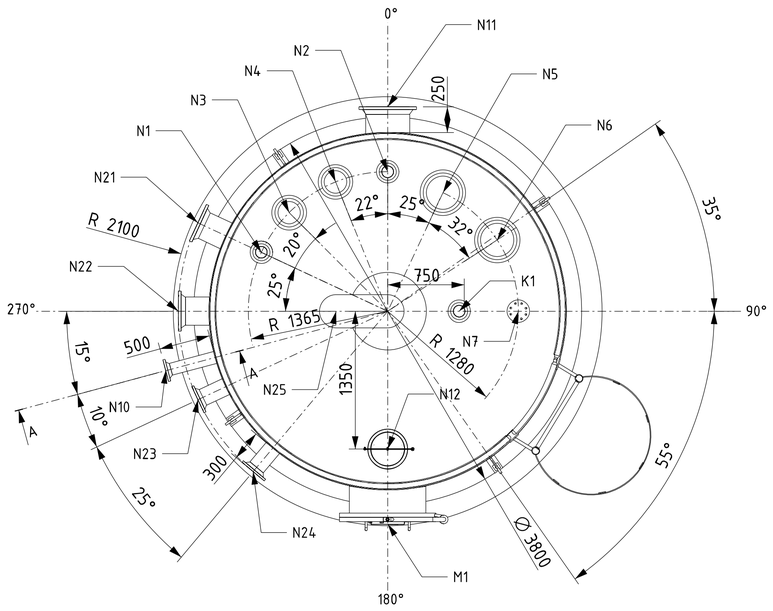
DRAWING NUMBER MD-216-3350-EG-ME-GAD-1027 REV A01



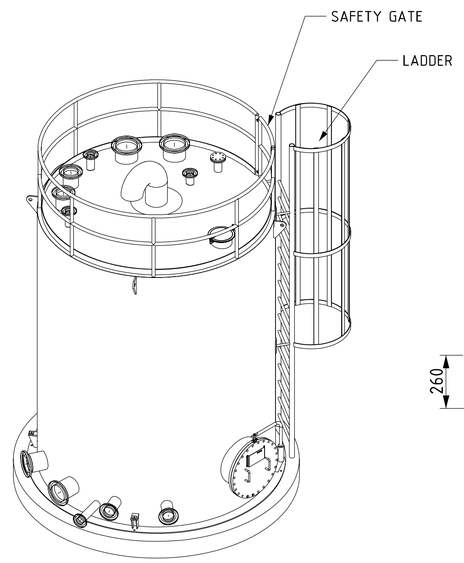
FRONT VIEW



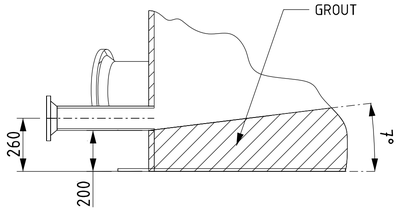
LEFT VIEW



TOP VIEW



ISOMETRIC VIEW 1:45



SECTION A-A 1:15

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3350-TNK-001
DESCRIPTION	FILTRATE TANK
DESIGN CODE	API 650
DESIGN TEMPERATURE	+5-45°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	0.977 kPa/ SBC 301,2007
CORROSION ALLOWANCE: SHELL INSIDE	2 mm
CORROSION ALLOWANCE: BOTTOM	2 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	5600mmx4780mmx4250mm
SHELL HEIGHT	4500mm
CAPACITY: TOTAL VOLUME	39m³
CAPACITY: LIVE VOLUME	22m³
SEISMIC DESIGN /CODE	MD-216-1000-EG-GE-SPC-1001 /SBC 301, 2007

LOAD DATA

TOTAL WEIGHT (FULL)	46100 kg
TANK WEIGHT	7100 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284/ EN10025-2 S235JR
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A284
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N1	4"	150	295°	250	-	4800	ASME B16.5	INLET, PROCESS WATER
N2	4"	150	0°	250	-	4800	ASME B16.5	INLET, CORE FLUSH DRAIN
N3	8"	150	315°	250	-	4800	ASME B16.5	INLET, FILTRATE
N4	8"	150	338°	250	-	4800	ASME B16.5	INLET, FILTRATE
N5	12"	150	25°	250	-	4800	ASME B16.5	INLET, CLOTH WASH DRAIN
N6	12"	150	57°	250	-	4800	ASME B16.5	INLET, CLOTH WASH DRAIN
N7	4"	150	90°	250	-	4800	ASME B16.5	SPARE
N21	10"	150	295°	300	-	400	ASME B16.5	OUTLET TO PUMP
N22	10"	150	270°	300	-	400	ASME B16.5	OUTLET TO PUMP
N23	6"	150	245°	300	-	400	ASME B16.5	OUTLET TO PUMP
N24	6"	150	220°	300	-	400	ASME B16.5	OUTLET TO PUMP
N25	12"	150	270°	-	-	5000	ASME B16.5	VENT
N10	4"	150	260°	500	-	260	ASME B16.5	DRAIN
N11	16"	150	0°	250	-	4100	ASME B16.5	EMERGENCY OVERFLOW
N12	12"	150	180°	-	-	4700	ASME B16.5	INSPECTION HOLE
M1	Ø30"	150	180°	-	-	850	ASME B16.5	MANHOLE
K1	4"	150	90°	-	-	4800	ASME B16.5	LEVEL INDICATOR

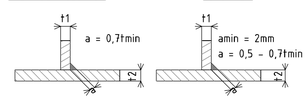
NOTE:
1. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
2. ALL ELEVATION MEASUREMENTS ARE MEASURED FROM BOTTOM SIDE OF BASE PLATE.
3. SHELL NOZZLE ELEVATIONS ARE FROM BOTTOM FROM BOTTOM OF BASE PLATE TO CENTERLINE OF NOZZLE.
4. SHELL NOZZLE PROJECTIONS ARE MEASURED FROM OUTSIDE OF SHELL TO FLANGE FACE.
5. ALL NOZZLE LOCATION, ELEVATION AND ORIENTATION ARE INDICATIVE ONLY AND WILL BE FINALIZED DURING DETAIL DESIGN PHASE.

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	31-01-20	ISSUED FOR APPROVAL	AV	TV	SC	RL	

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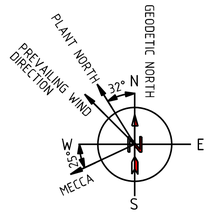
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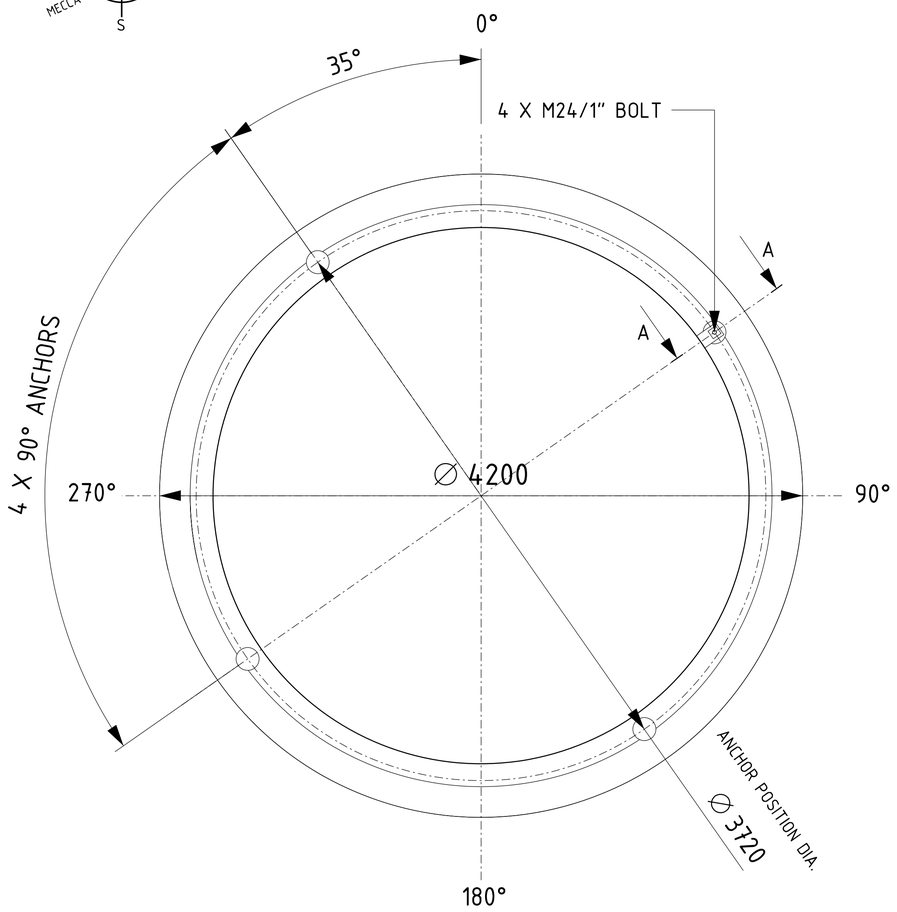
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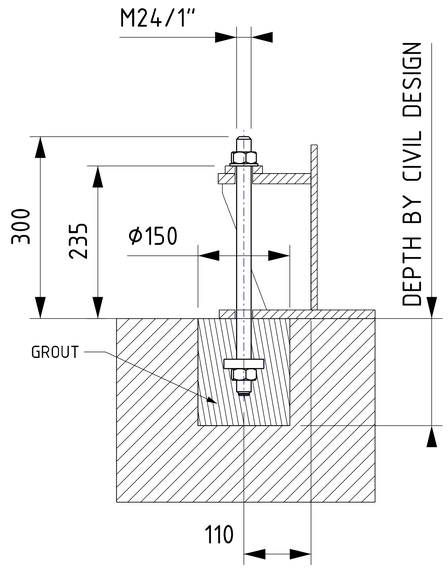
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OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707_GAU01_SF01_MTA01_33516		OU100661833	0
DRAWN	AliVai	31-01-20	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	TatVie	31-01-20	OUTLINE DRAWING
ENGINEER	SouCha	31-01-20	CONCENTRATE FILTRATION
APPROVED	RicLeh	31-01-20	3350-TNK-001 FILTRATE TANK
EMPLOYER			
SCALE 1:30	A1		DRAWING NUMBER MD-216-3350-EG-ME-GAD-1027 REV A01

DRAWING NUMBER MD-216-3350-EG-ME-GAD-1057 REV A01







SECTION A-A 1:5

FOOTING LOAD TABLE
3350-TNK-001 FILTRATE TANK

LOAD	DESCRIPTION	VERTICAL (kN)	HORIZONTAL (kN)	MOMENT (kNm)
1	TANK DEAD LOAD	70	0,0	0,0
2	NORMAL OPERATING FLUID LOAD SG_{opp} = 1,0 t/m³	384	0,0	0,0
3	MAX FLUID LOAD SG_{ha-ded}= 1,0 t/m³	452	0,0	0,0
4	SEISMIC LOAD	0,0	18	50
5	WIND LOAD	0,0	15	41

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
- S1= 0,02g
- Ss= 0,10g
- max. ground elevation 3m

CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43m/s

LOADING NOTES:
1. ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON GROUND OR CONCRETE SLAB)

FOUNDATION AND SUPPORT TOLERANCES:
1. THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2. THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL

CONCRETE NOTES: (WHERE TANKS ARE INSTALLED ON CONCRETE)
1. CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2. FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWN BOLT EDGE DISTANCE TO BE DETERMINED BY CIVIL ENGINEERING

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	31-01-20	ISSUED FOR APPROVAL	AV	TV	SC	RL	

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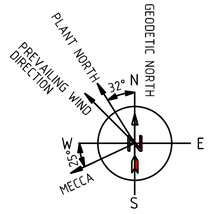
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OUTOTEC DRAWING NUMBER: 900707_GAU01_SF01_MLD15 _33502 — DOCUMENT ID: OU100671680 — REVISION: 0

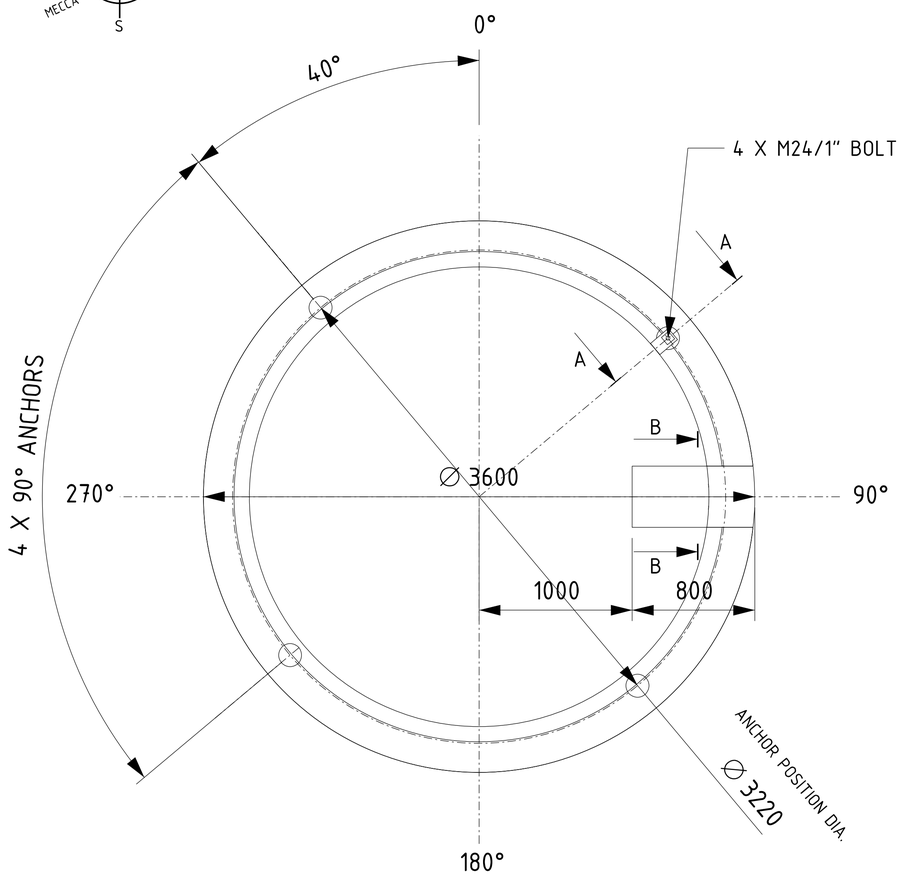
DRAWN	AliVai	31-01-20
CHECKED	TatVie	31-01-20
ENGINEER	SouCha	31-01-20
APPROVED	RicLeh	31-01-20
EMPLOYER		
SCALE	1:20	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
CONCENTRATE FILTRATION
3350-TNK-001 FILTRATE TANK

DRAWING NUMBER: MD-216-3350-EG-ME-GAD-1057 — REV: A01

DRAWING NUMBER MD-216-3350-EG-ME-GAD-1058 REV A01





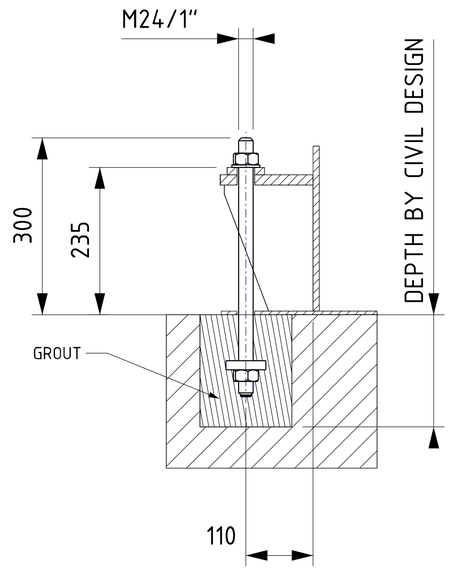
FOOTING LOAD TABLE
3350-TNK-001 FILTRATE TANK

LOAD	DESCRIPTION	VERTICAL (kN)	HORIZONTAL (kN)	MOMENT (kNm)
1	TANK DEAD LOAD	48	0,0	0,0
2	NORMAL OPERATING FLUID LOAD SG_{liq} = 1,0 t/m³	244	0,0	0,0
3	MAX FLUID LOAD $SG_{max liq}$ = 1,0 t/m³	323	0,0	0,0
4	SEISMIC LOAD SEISMIC, WIND DESIGN /CODE: MD-216-1000-EG-GE-SPC-1001 /SBC 301, 2007	0,0	13	32
5	WIND LOAD SEISMIC, WIND DESIGN /CODE: MD-216-1000-EG-GE-SPC-1001 /SBC 301, 2007	0,0	11	29

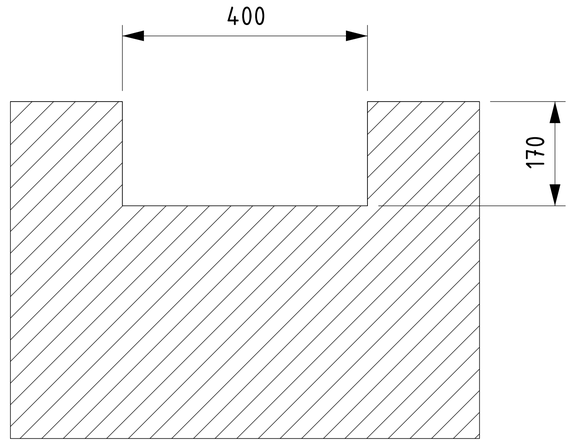
SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
- S1= 0,02g
- Ss= 0,10g
- max. ground elevation 3m

CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43m/s



SECTION A-A 1:5



SECTION B-B 1:5

SPACE FOR DRAIN PIPE

LOADING NOTES:
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OUTOTEC DRAWING NUMBER	900707_GAU01_SF01_MLD15 _33503	DOCUMENT ID OU100671681	REVISION 0

DRAWN	AliVai	31-01-20
CHECKED	TatVie	31-01-20
ENGINEER	SouCha	31-01-20
APPROVED	RicLeh	31-01-20
EMPLOYER		
SCALE	1:20	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
CONCENTRATE FILTRATION
3350-TNK-002 FILTER CLOTH WASH TANK

DRAWING NUMBER MD-216-3350-EG-ME-GAD-1058 REV A01